测试报告

测试总结果：NG

SN：010530079S3030011

测试时间：20260318_101028

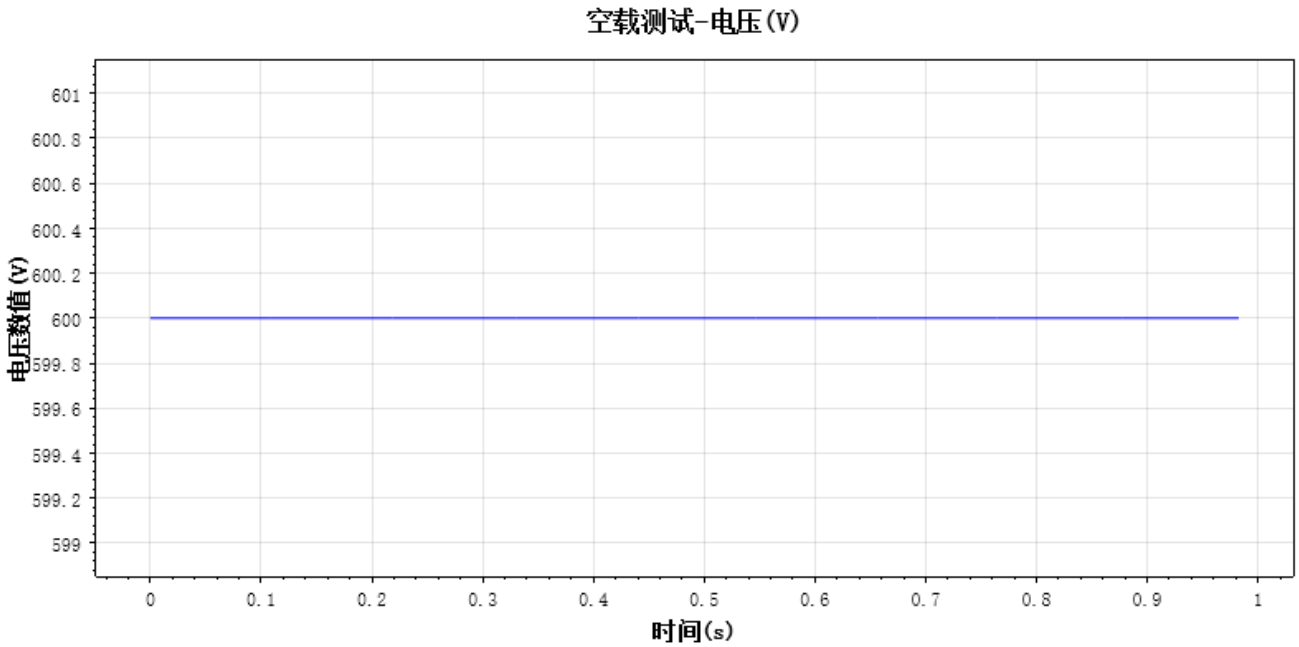
测试工位：压缩机氮气台架_1#

空载测试

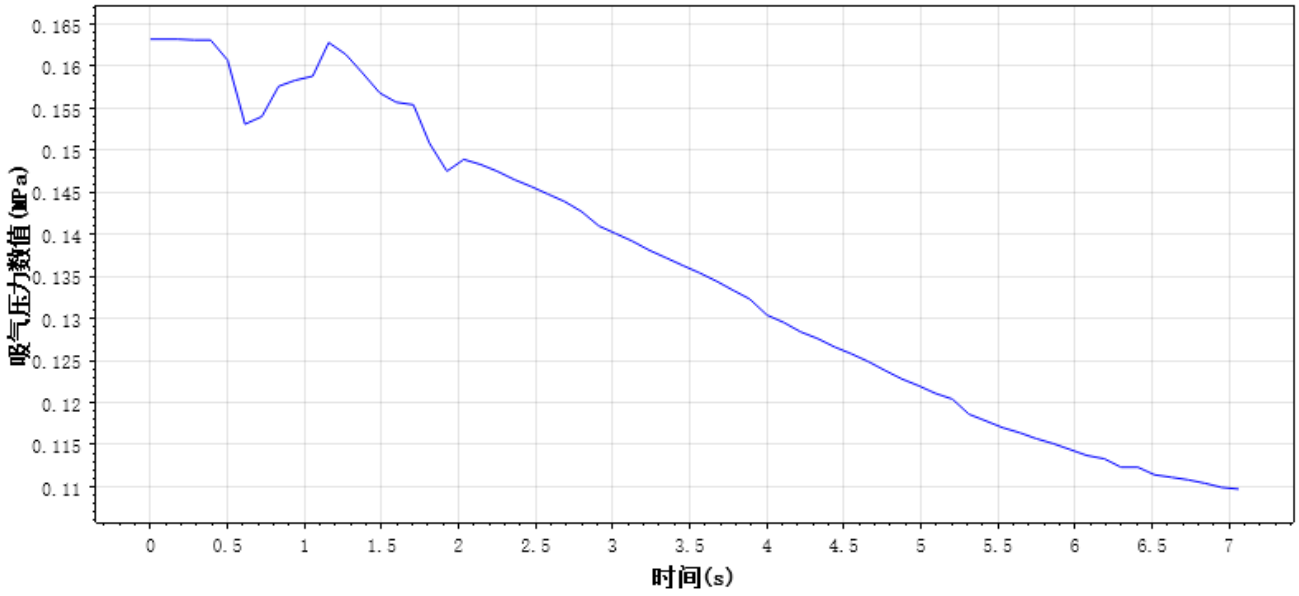
测试结果：OK

空载测试-单值数据

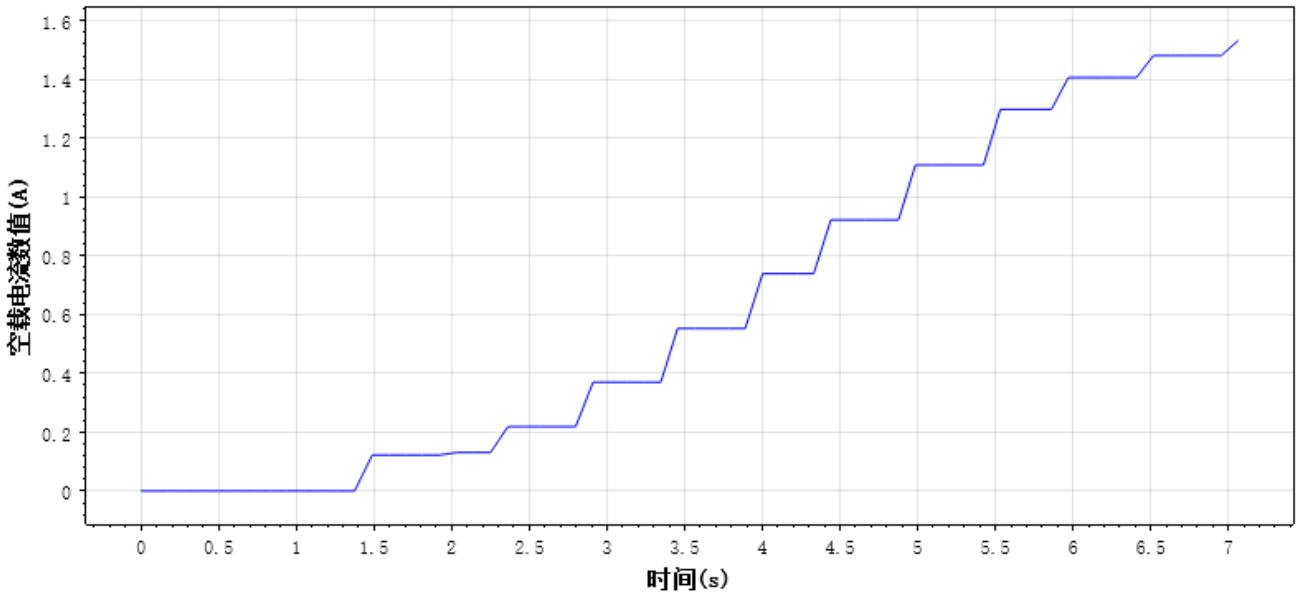
名称	单位	数据	下限	上限	结果
电压(V)	V	600	585	615	OK
吸气压力(MPa)	MPa	0.11	0.10	0.12	OK
空载电流(A)	A	1.53	0	4	OK
空载功率(W)	W	918.84	0	1150	OK
空载流量(SLPM)	SLPM	186.25	180	220	OK
排气压力(MPa)	MPa	0.11	0	1.20	OK
空载时间(s)	s	7.07	0	60	OK



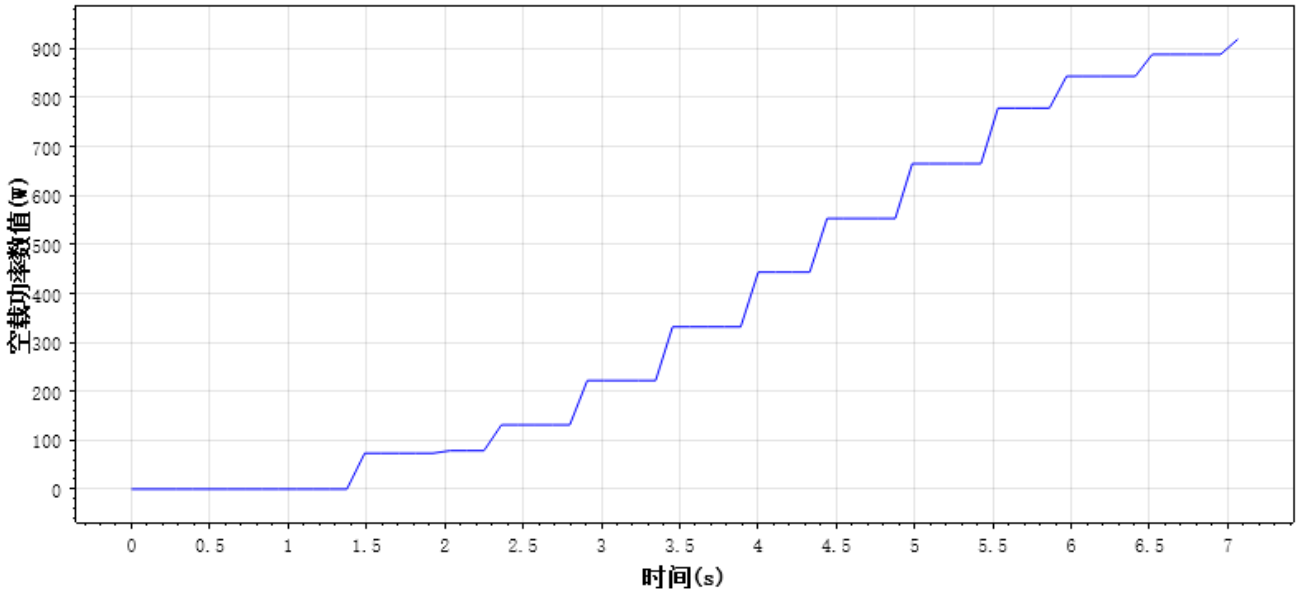
空载测试-吸气压力(MPa)



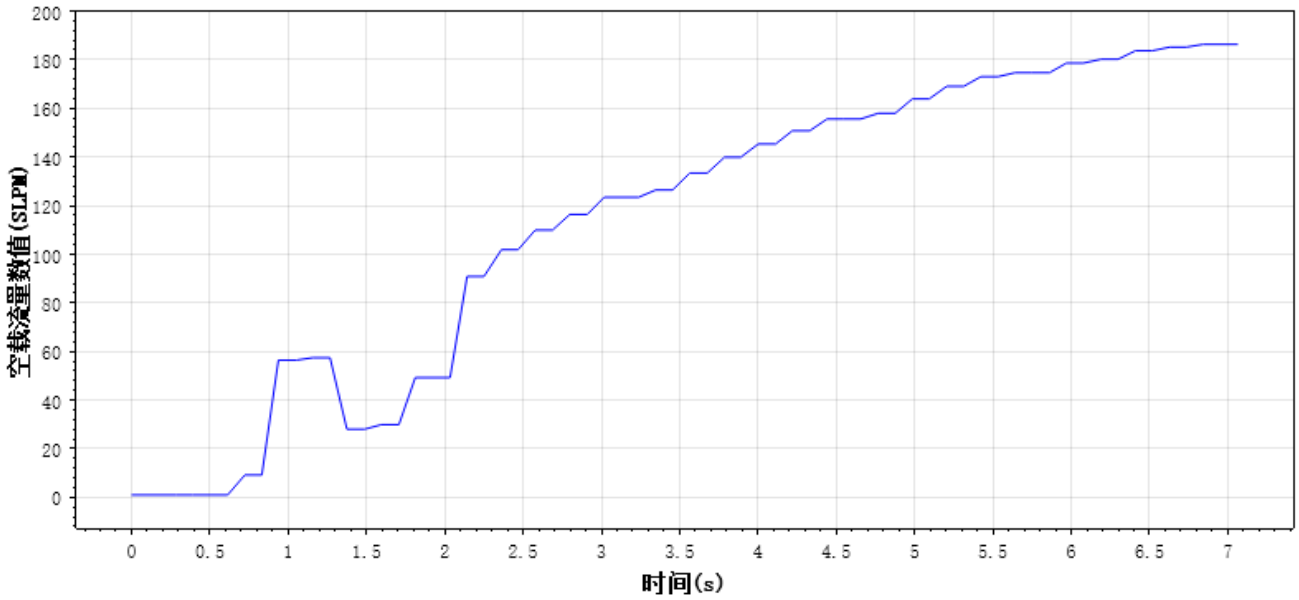
空载测试-空载电流(A)

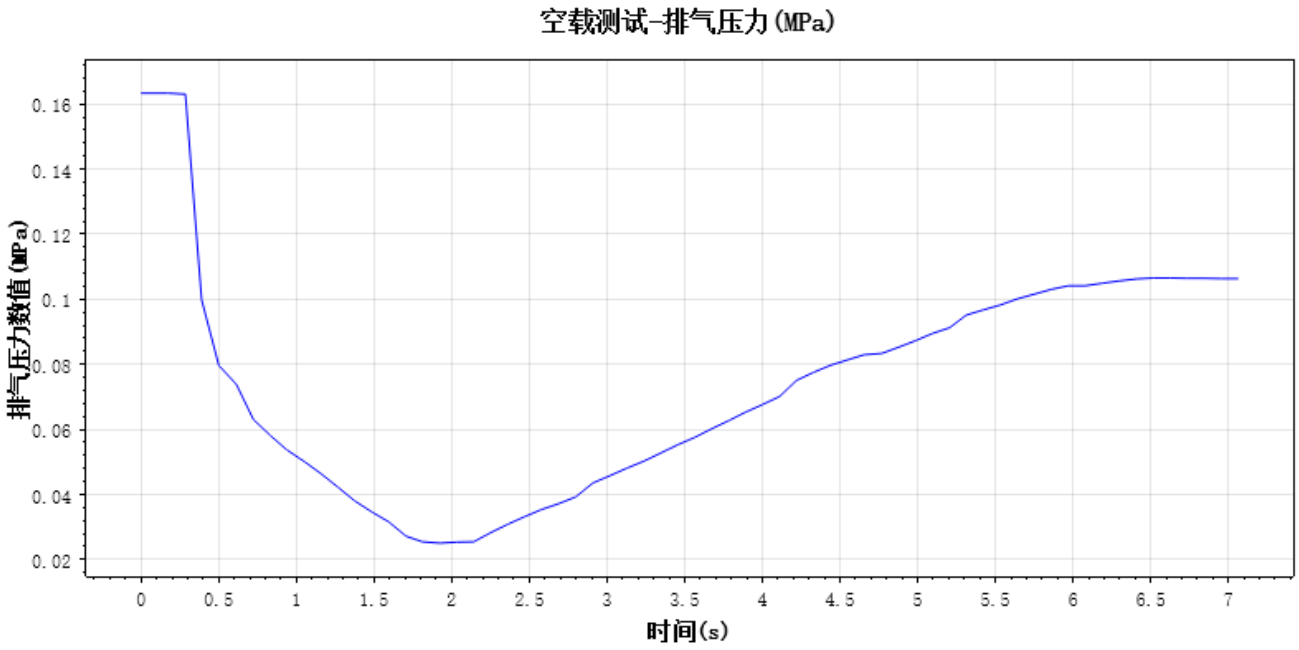


空载测试-空载功率(W)



空载测试-空载流量(SLPM)





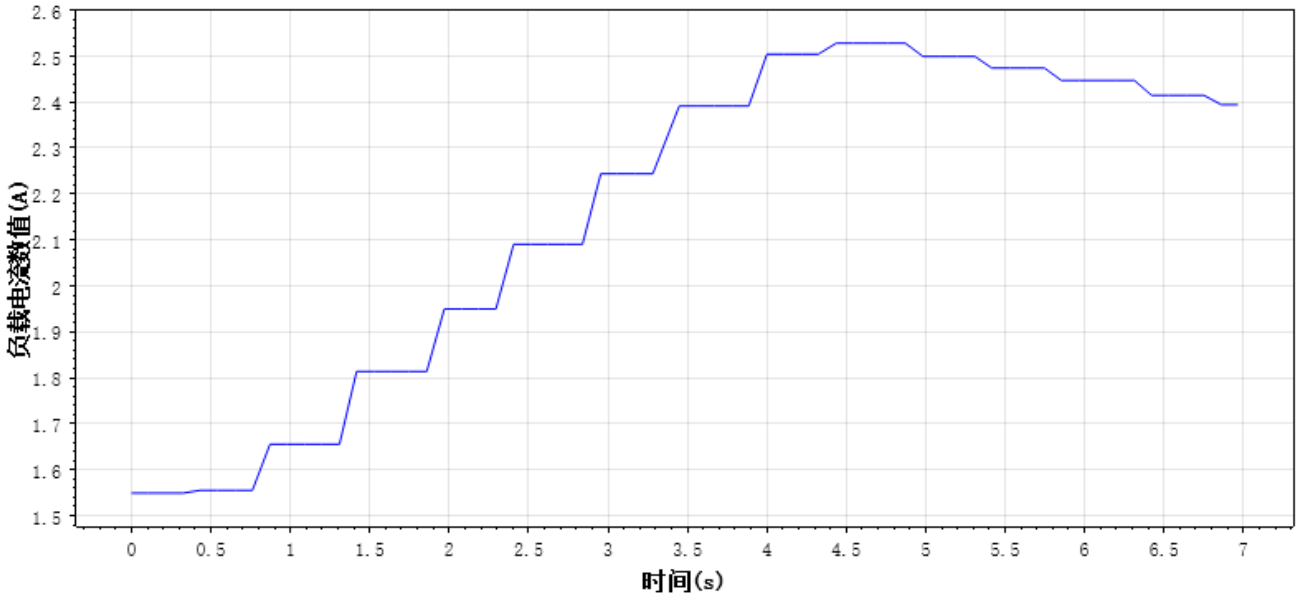
负载测试

测试结果：OK

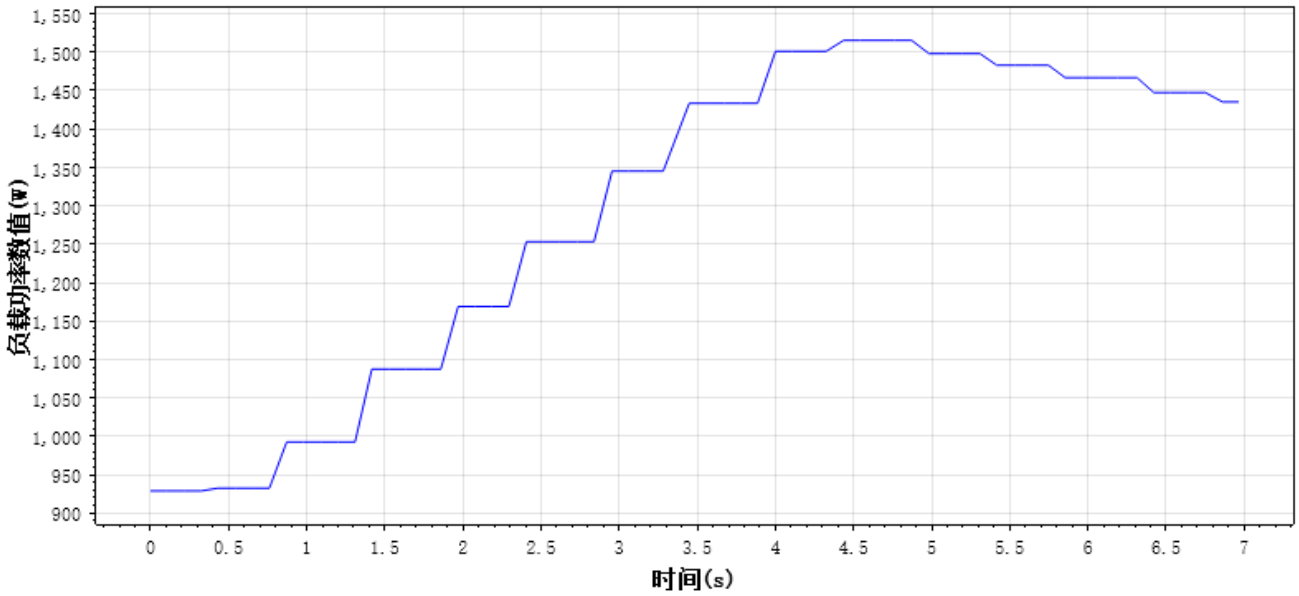
负载测试-单值数据

名称	单位	数据	下限	上限	结果
升压时间(s)	s	3.64	0	5.50	OK
负载电流(A)	A	2.39	0	6.50	OK
负载功率(W)	W	1435.13	0	2200	OK
负载流量(SLPM)	SLPM	191.26	180	220	OK
升压压力(MPa)	MPa	1.19	1.10	1.30	OK
逆流压降(MPa)	MPa	0.0046	0	0.050	OK

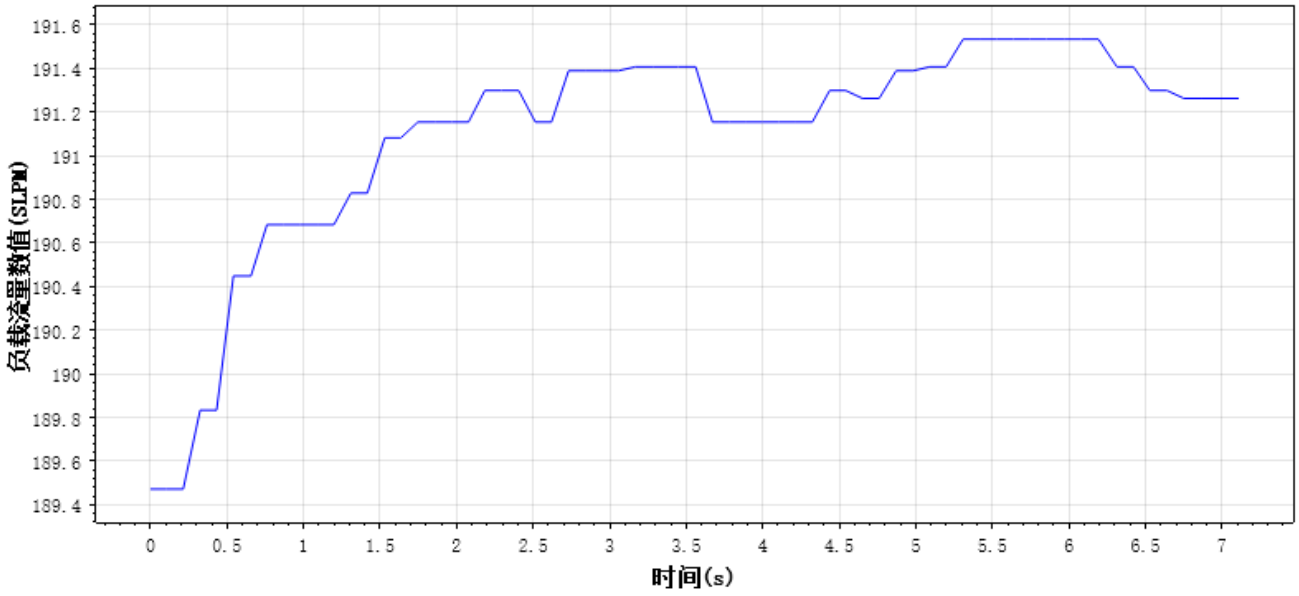
负载测试-负载电流(A)



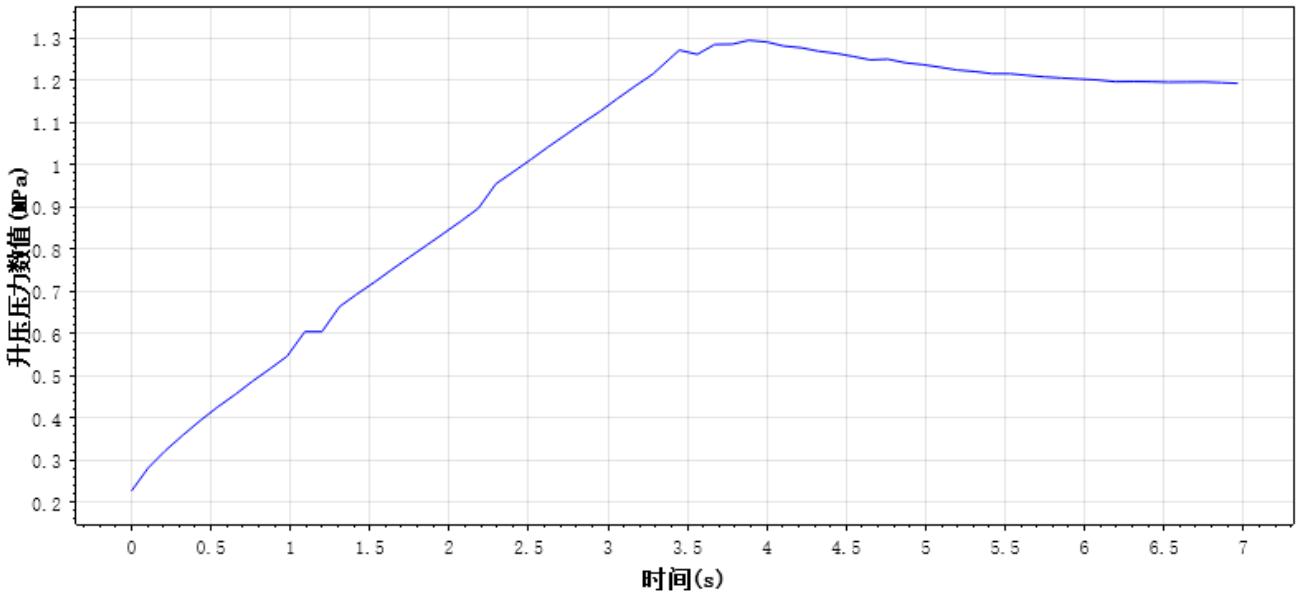
负载测试-负载功率(W)

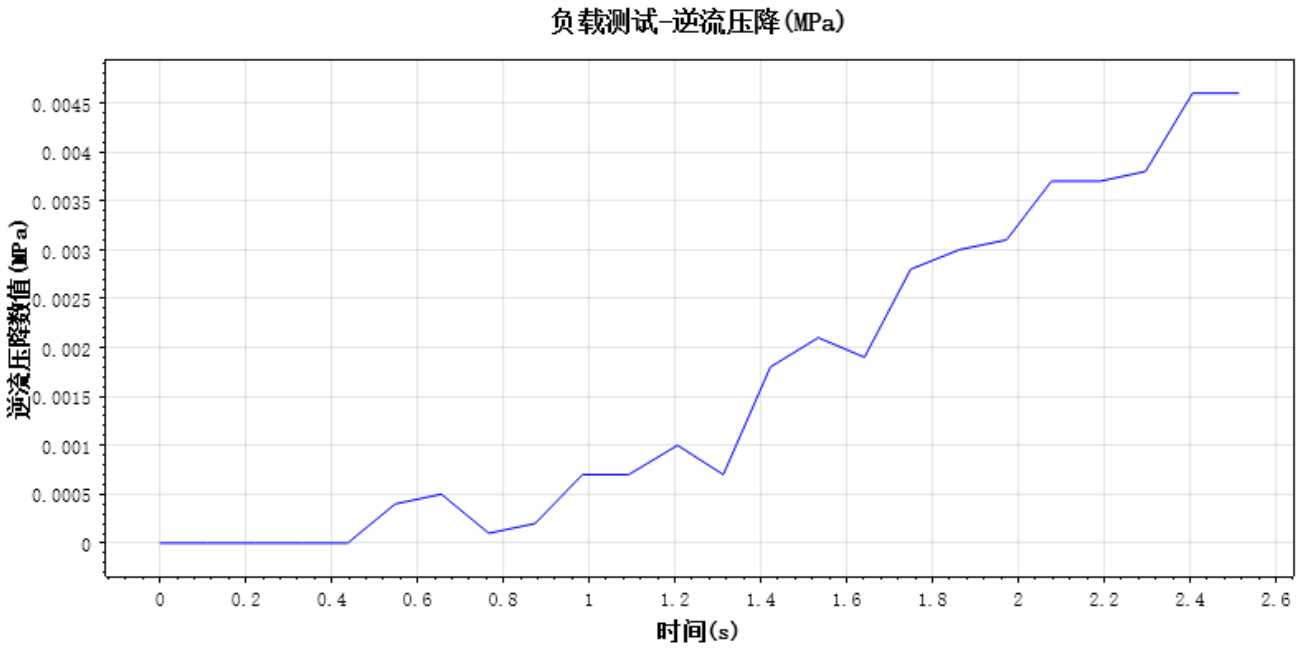


负载测试-负载流量(SLPM)



负载测试-升压压力(MPa)





NVH工况1

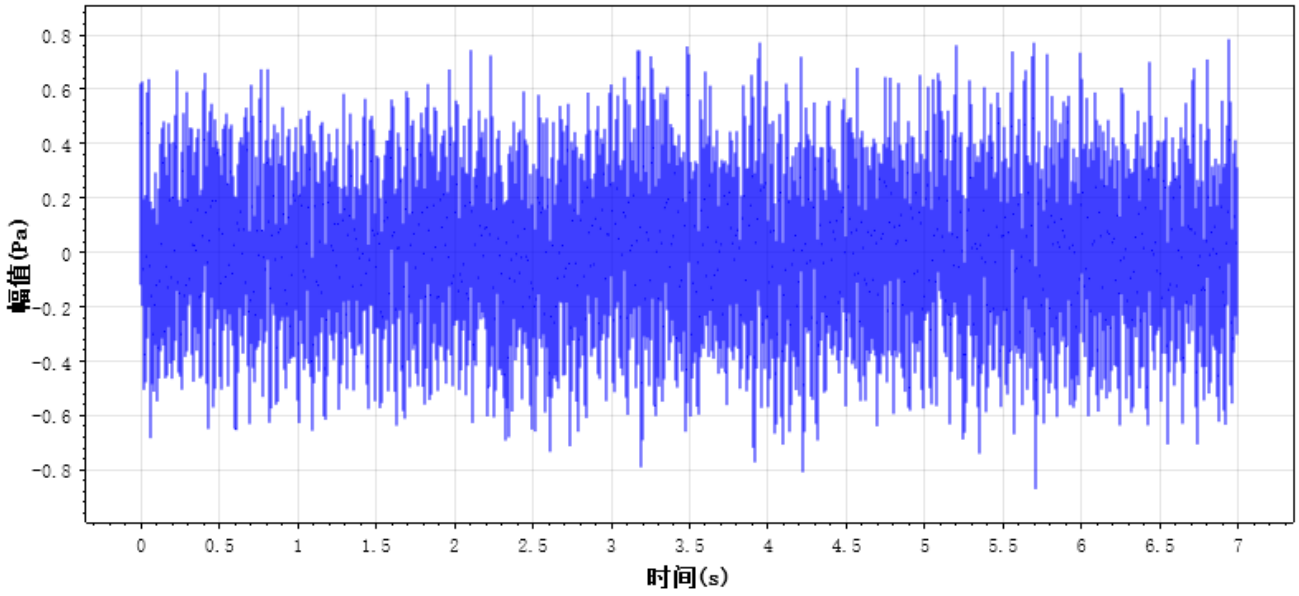
测试结果：NG

工况信息：转速-800rpm； 进气压力-0.3MPa； 排气压力-1MPa

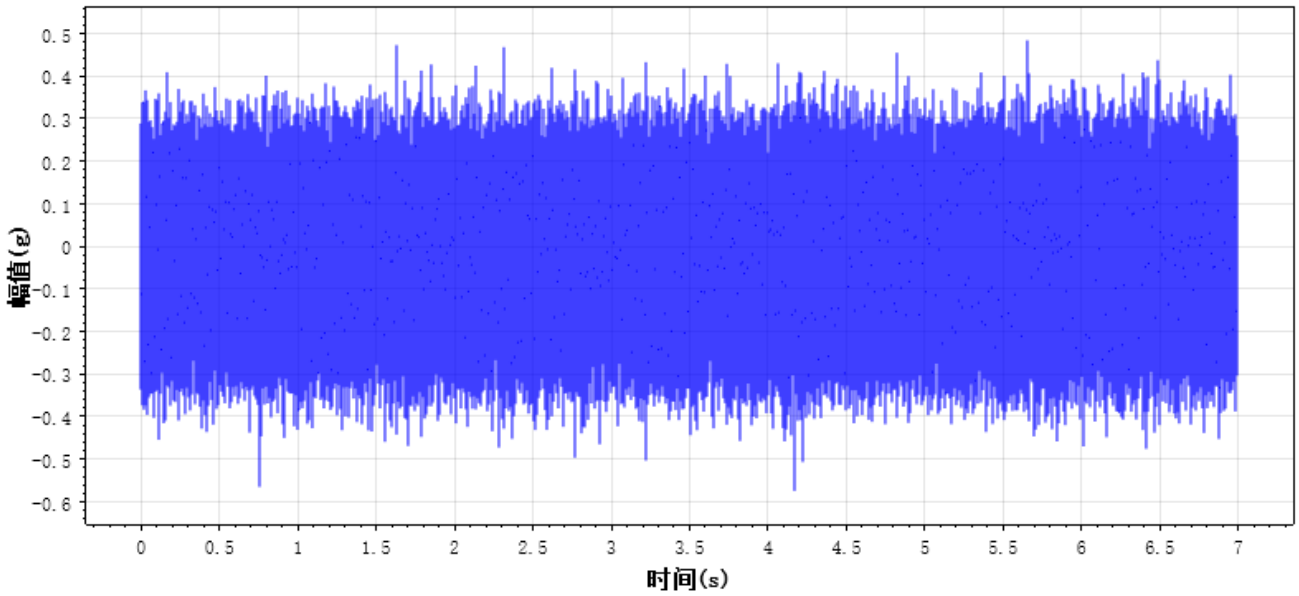
NVH工况1-单值数据

名称	单位	数据	下限	上限	结果
小波变换切片-主轴承Acc2-均值(多值)	g	646.94	0	0.012	NG
振动-功率谱-主轴承Acc2-均值(1000~1500Hz)	g	0.00062	0.00010	0.0060	OK
倍频程-主轴承Acc2-均值(1000~4000)	dB	50.19	0	72	OK
单点判定-倍频程-主轴承Acc2	dB	57.31	10	80	OK
时变响度-主轴承Acc2-均值(1~3s)	Sone	19.96	1	38	OK
特征响度-主轴承Acc2-均值(8~18)	Sone/Bark	0.76	0	3	OK
粗糙度-时间-主轴承Acc2-均值(3~6)	asper	0.12	0	0.80	OK
粗糙度-频率-主轴承Acc2-均值(2000~4000)	Asper/Bark	0.0047	0	0.050	OK

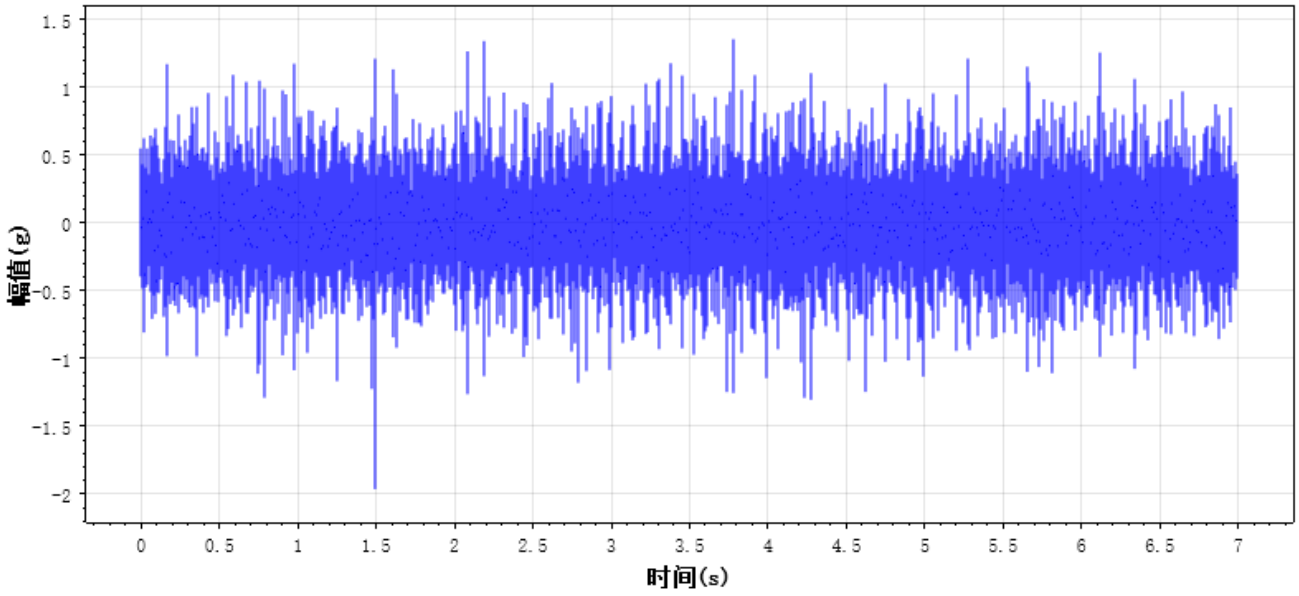
NVH工况1-多采集卡采集NVH数据-顶部Z向Mic1



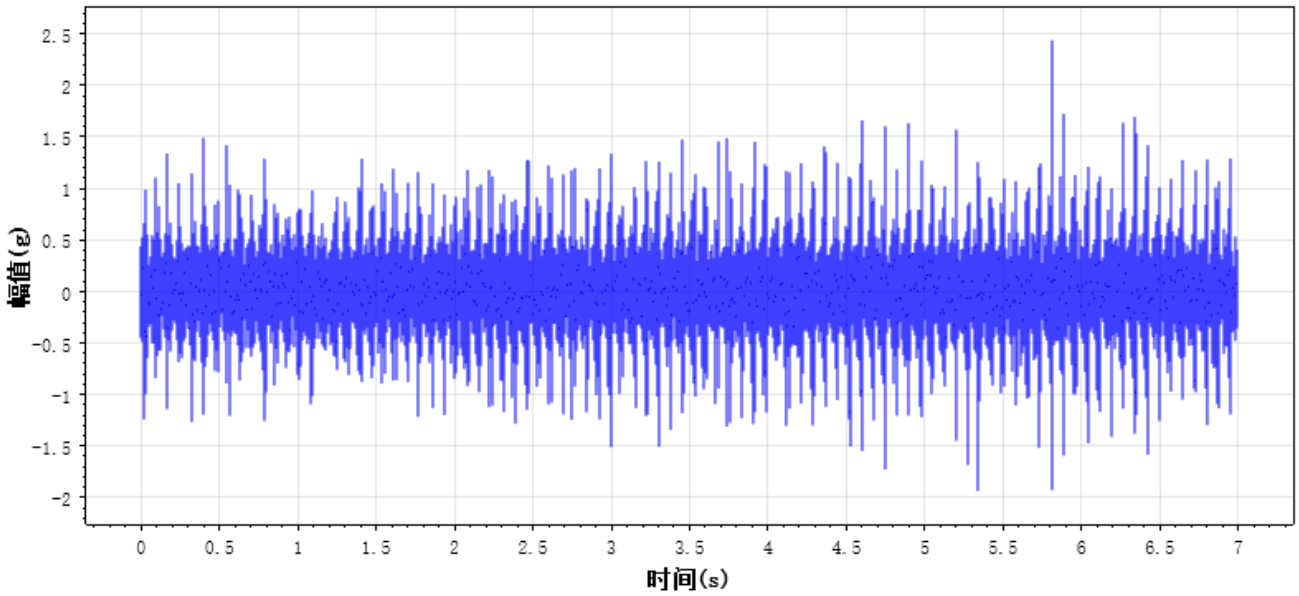
NVH工况1-多采集卡采集NVH数据-副轴承Acc1



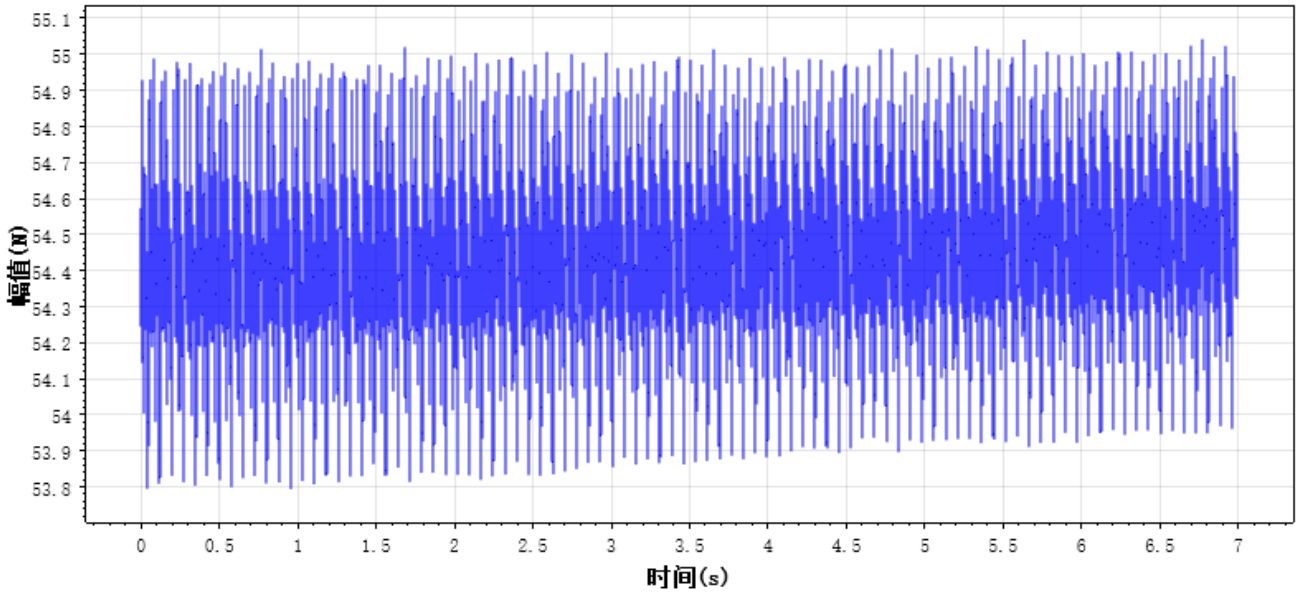
NVH工况1-多采集卡采集NVH数据-主轴承Acc2



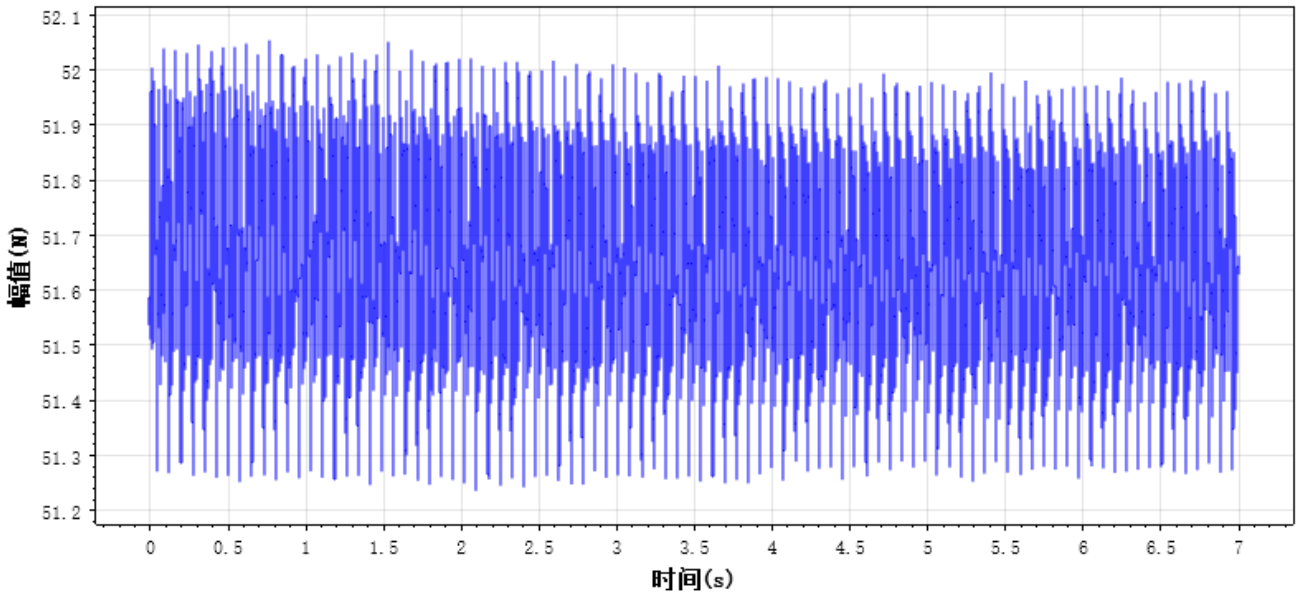
NVH工况1-多采集卡采集NVH数据-涡旋盘Acc3



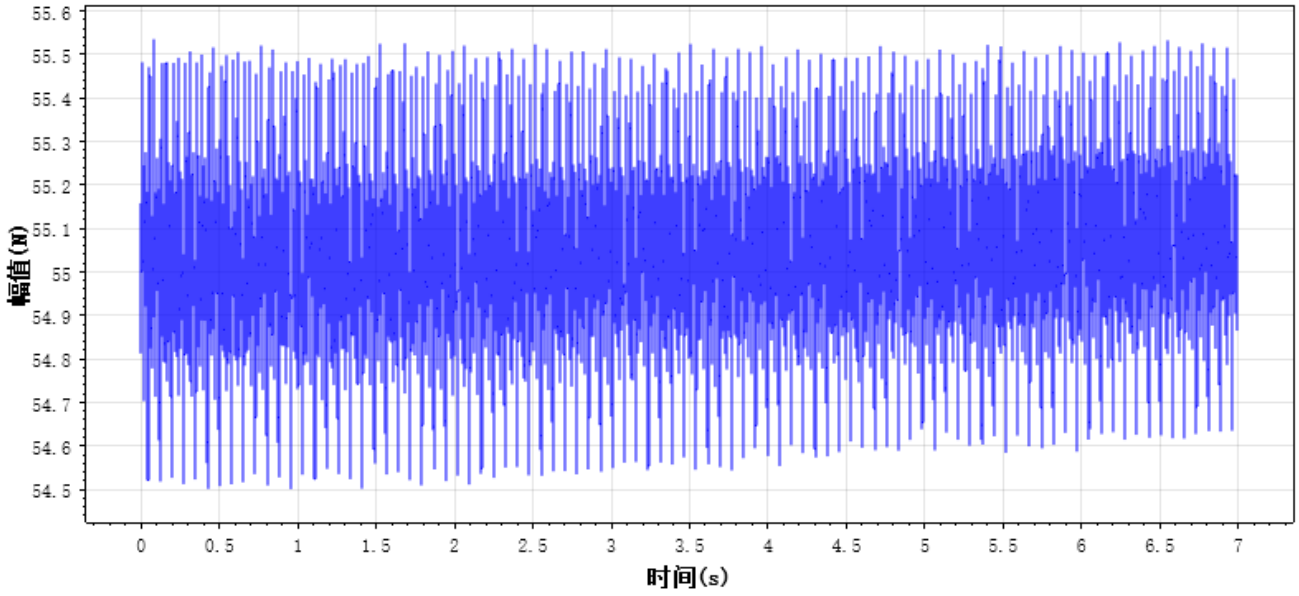
NVH工况1-多采集卡采集NVH数据-SS824-力馈1



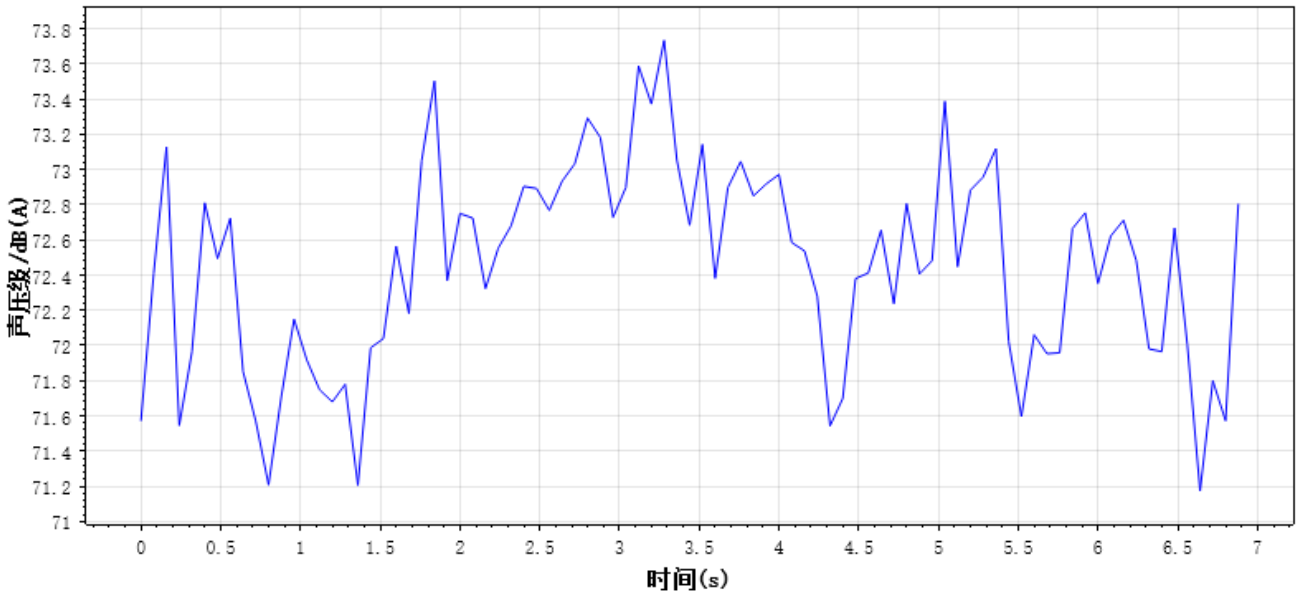
NVH工况1-多采集卡采集NVH数据-SS824-力馈2

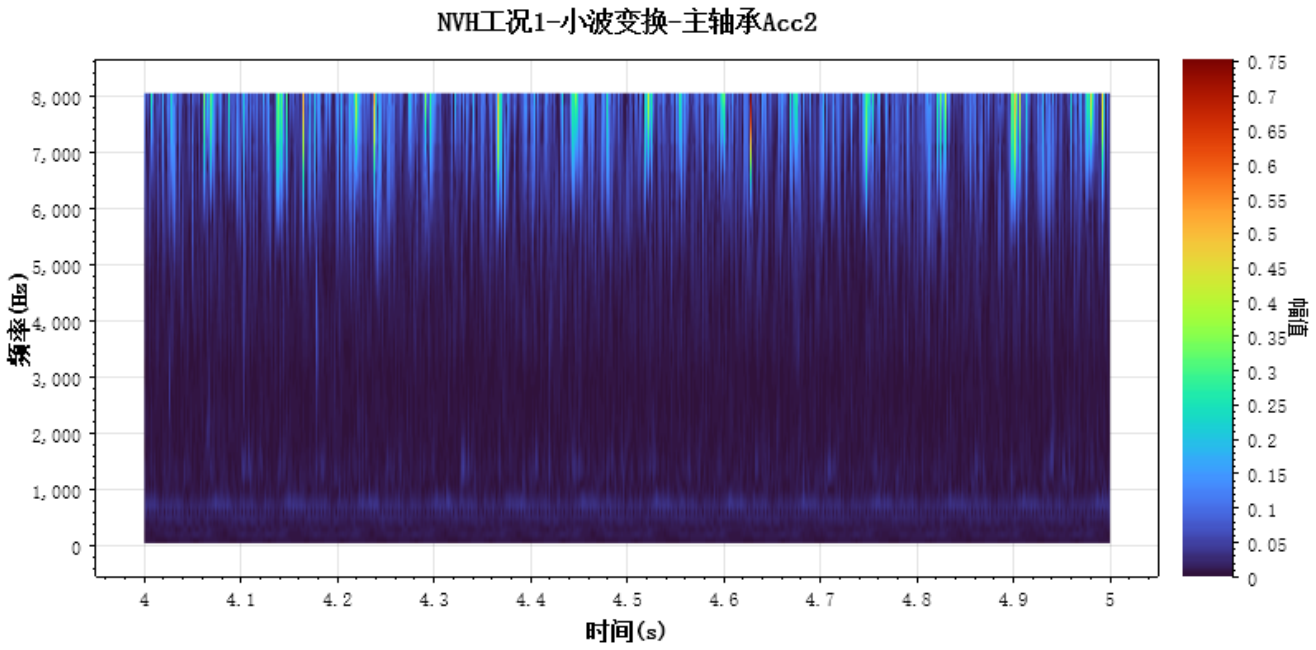


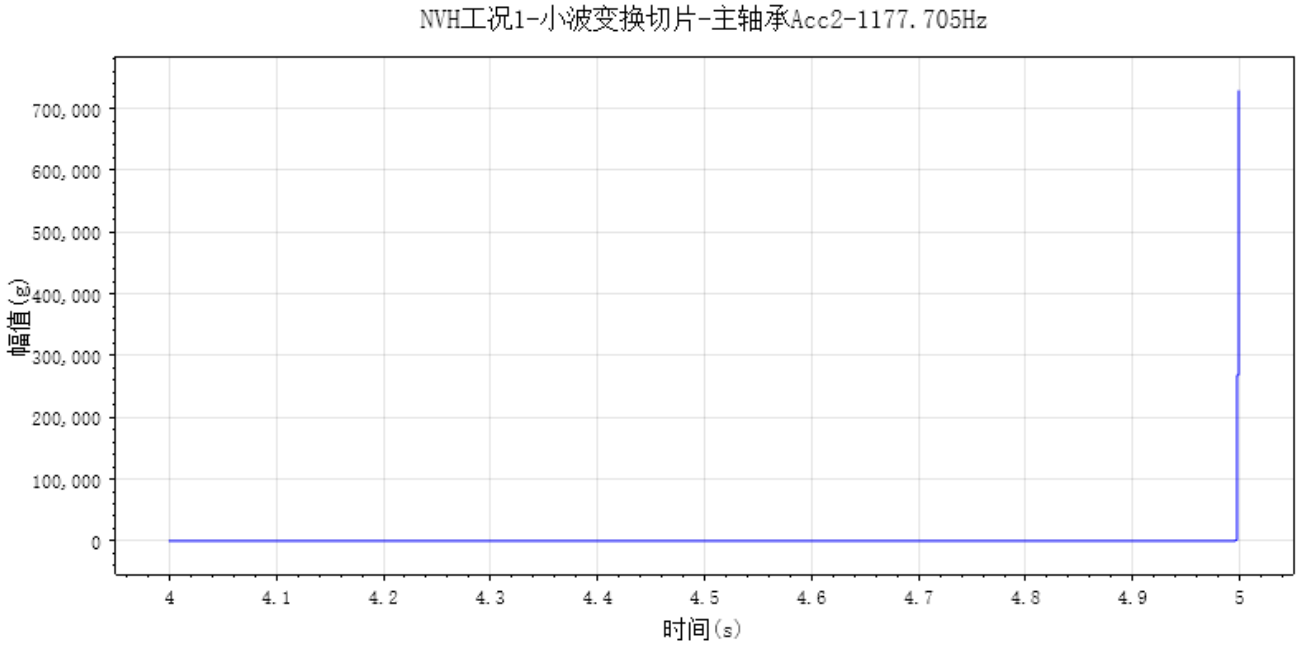
NVH工况1-多采集卡采集NVH数据-SS824-力馈3



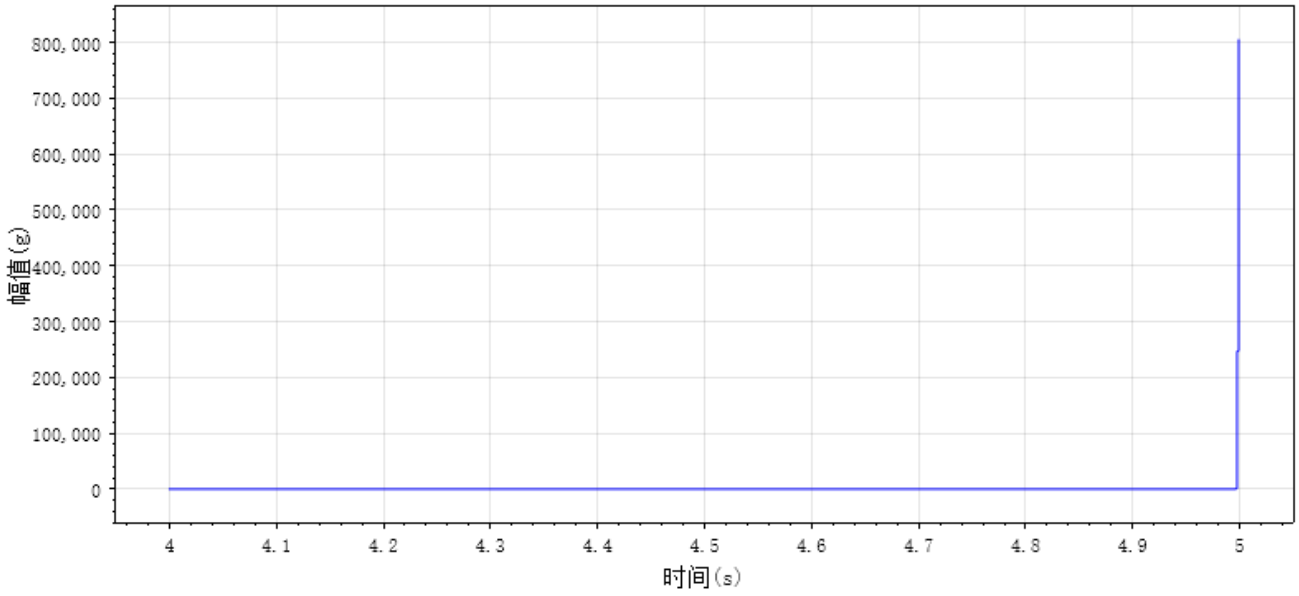
NVH工况1-声压级-顶部Z向Mic1



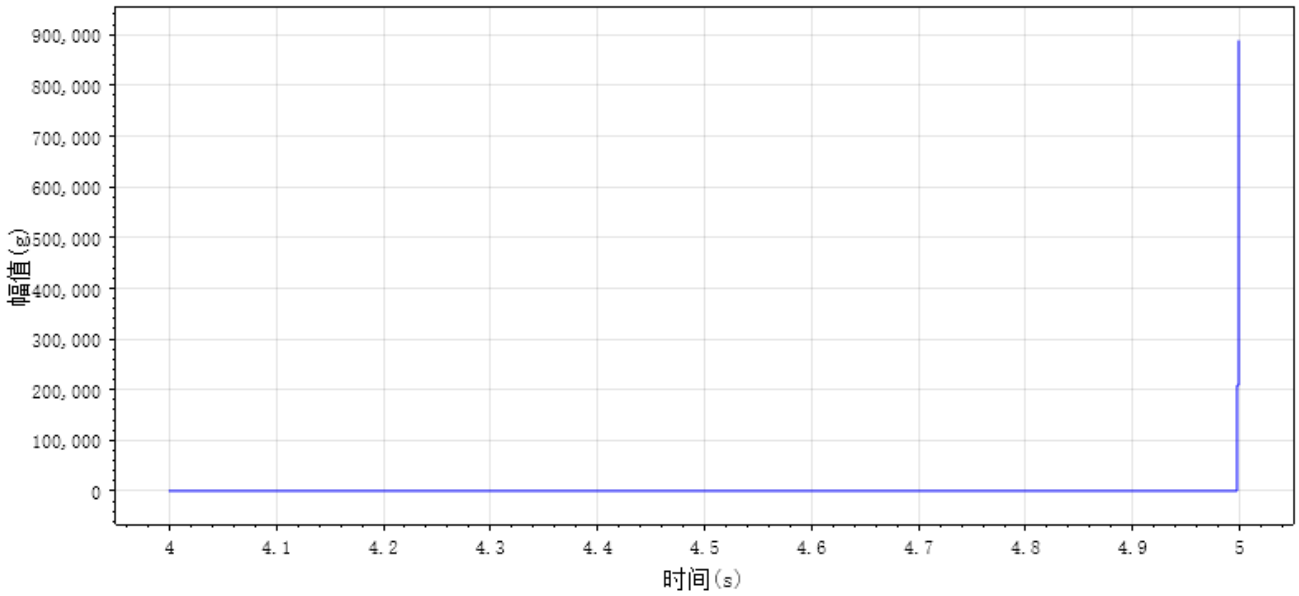

NVH工况1-小波变换-主轴承Acc2
频率(Hz)
8,000
7,000
6,000
5,000
4,000
3,000
2,000
1,000
0
4
4.1
4.2
4.3
4.4
4.5
4.6
4.7
4.8
4.9
5
时间(s)
幅值
0.75
0.7
0.65
0.6
0.55
0.5
0.45
0.4
0.35
0.3
0.25
0.2
0.15
0.1
0.05
0


NVH工况1-小波变换切片-主轴承Acc2-1177.705Hz
幅值(g)
700,000
600,000
500,000
400,000
300,000
200,000
100,000
0
4
4.1
4.2
4.3
4.4
4.5
4.6
4.7
4.8
4.9
5
时间(s)

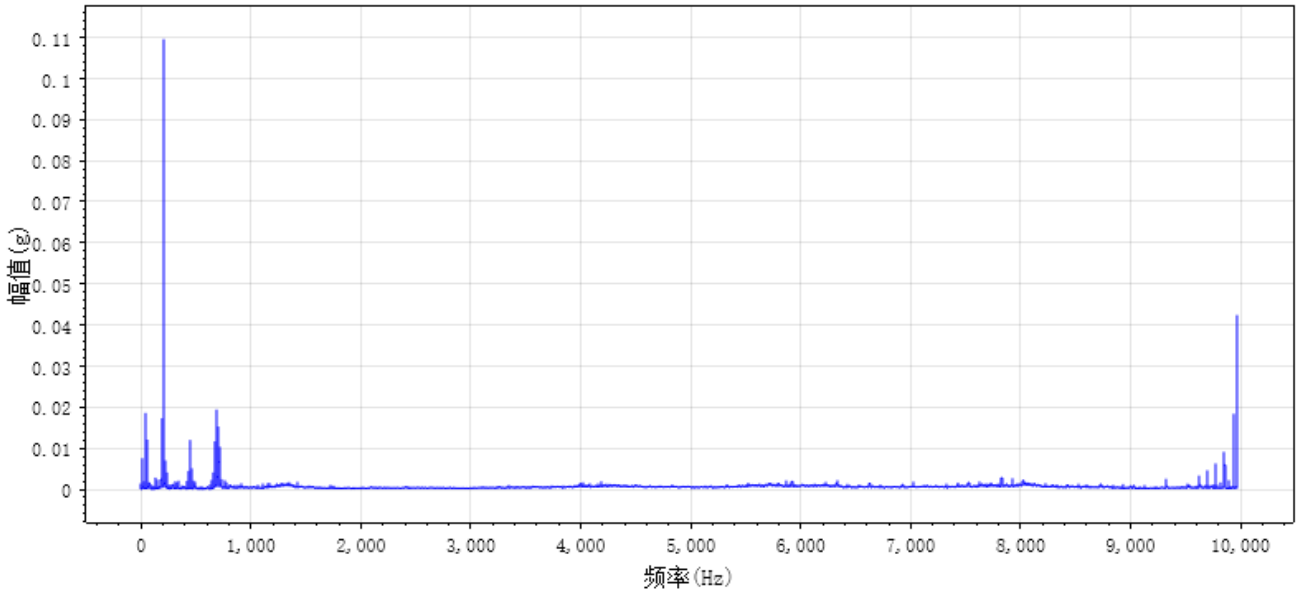
NVH工况1-小波变换切片-主轴承Acc2-1284.296Hz



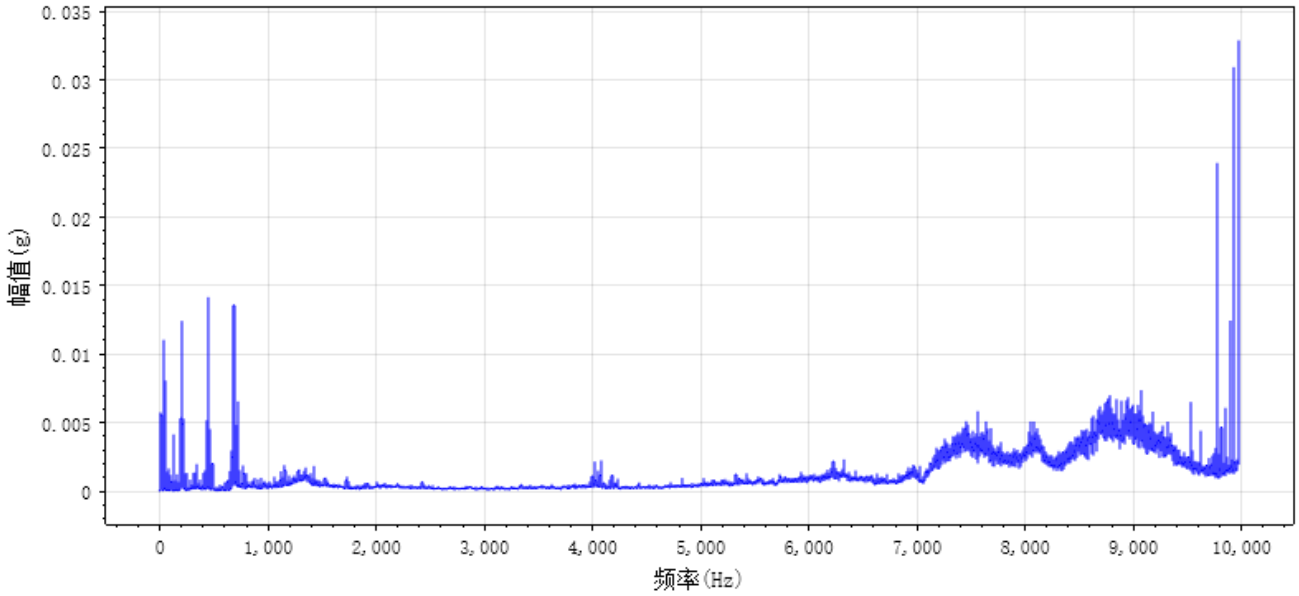
NVH工况1-小波变换切片-主轴承Acc2-1400.535Hz



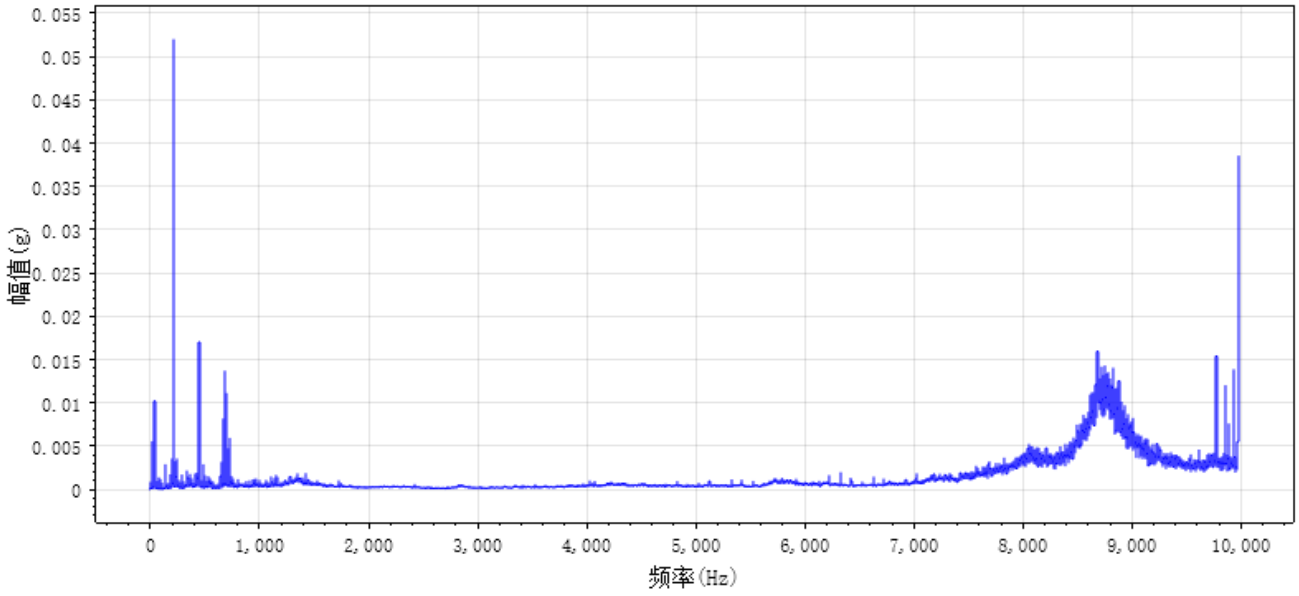
NVH工况1-振动-功率谱-副轴承Acc1



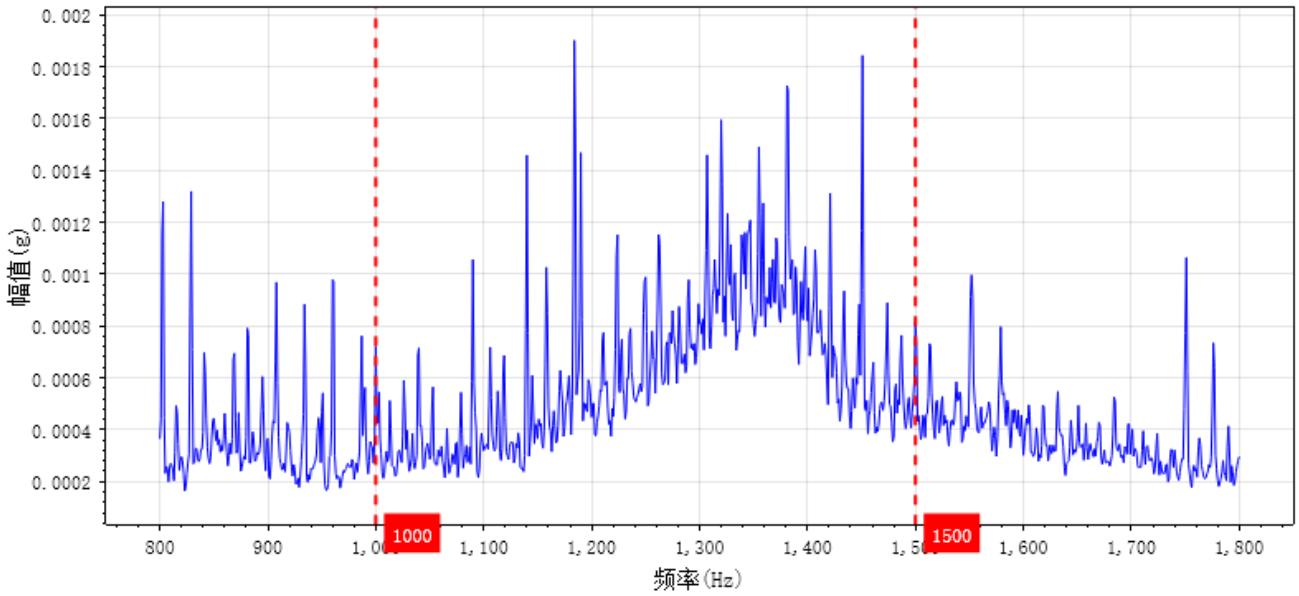
NVH工况1-振动-功率谱-主轴承Acc2



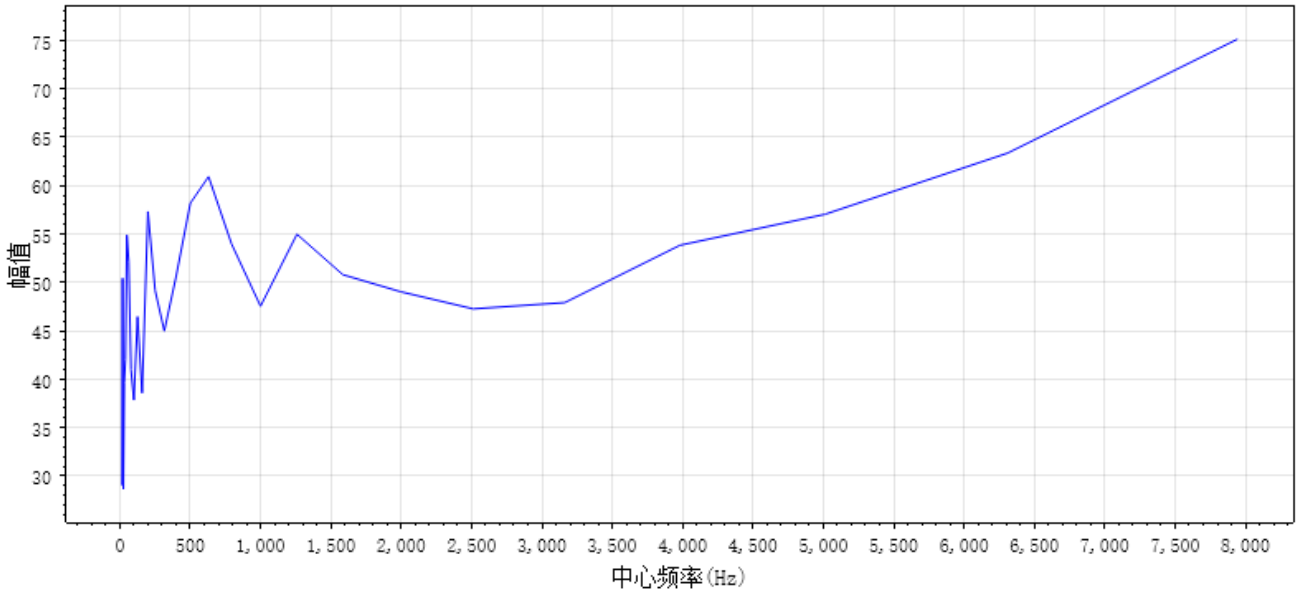
NVH工况1-振动-功率谱-涡旋盘Acc3



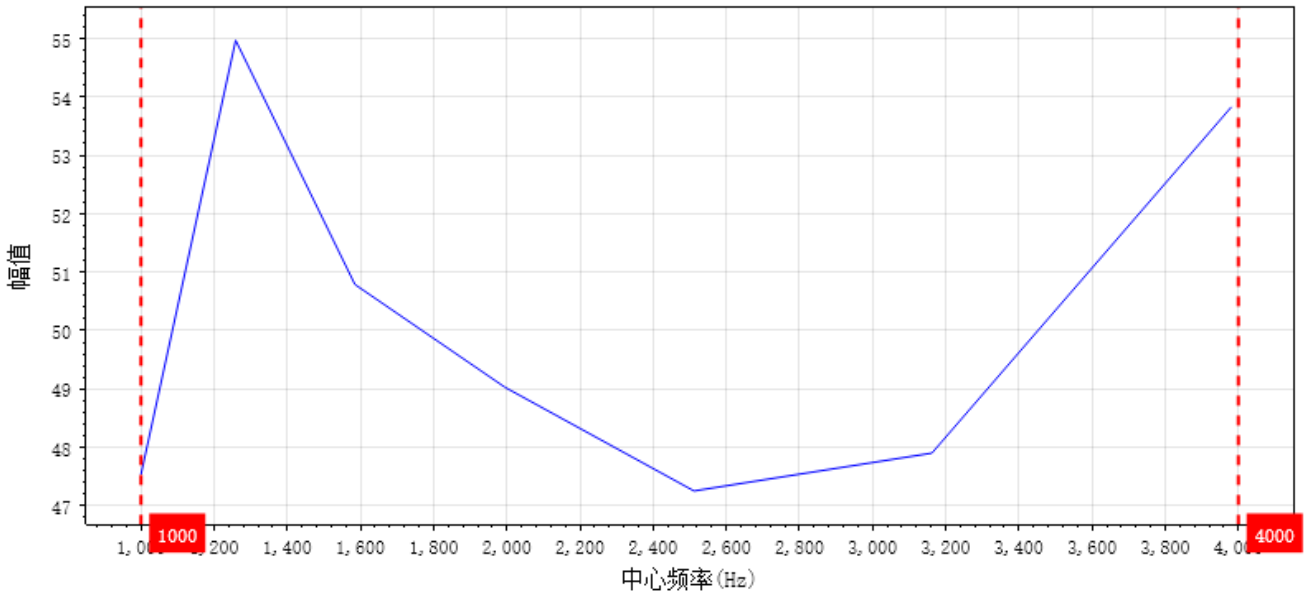
NVH工况1-振动-功率谱-主轴承Acc2-均值(0.00062)



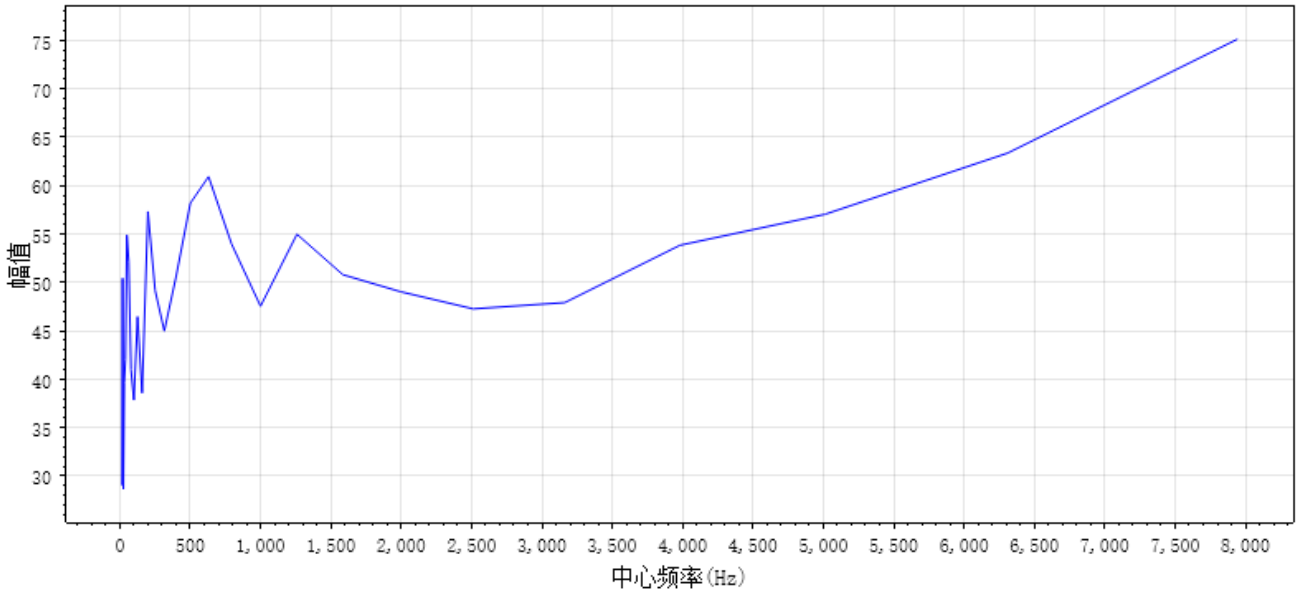
NVH工况1-倍频程-主轴承Acc2



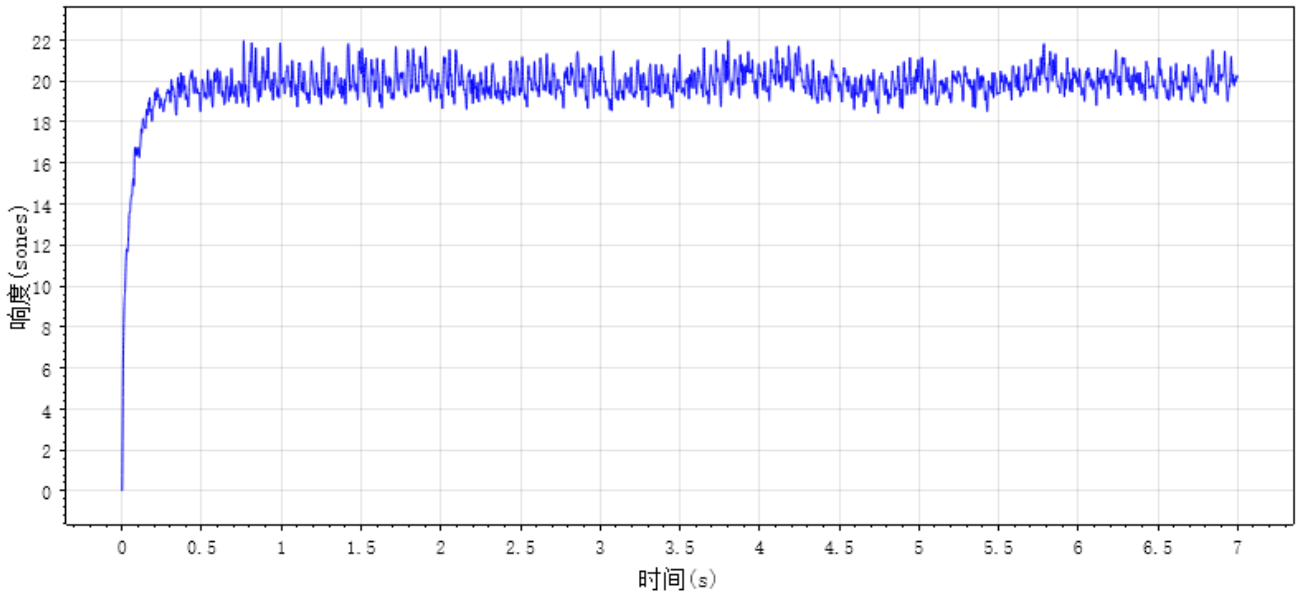
NVH工况1-倍频程-主轴承Acc2-均值(50.19)



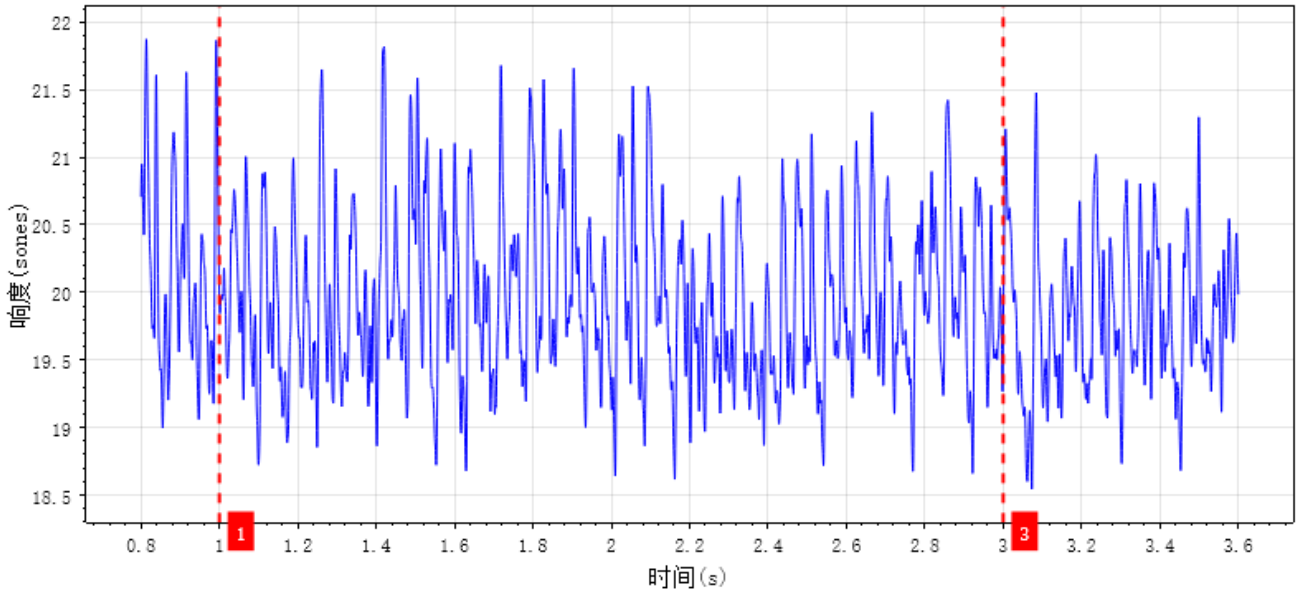
NVH工况1-单点判定-倍频程-主轴承Acc2



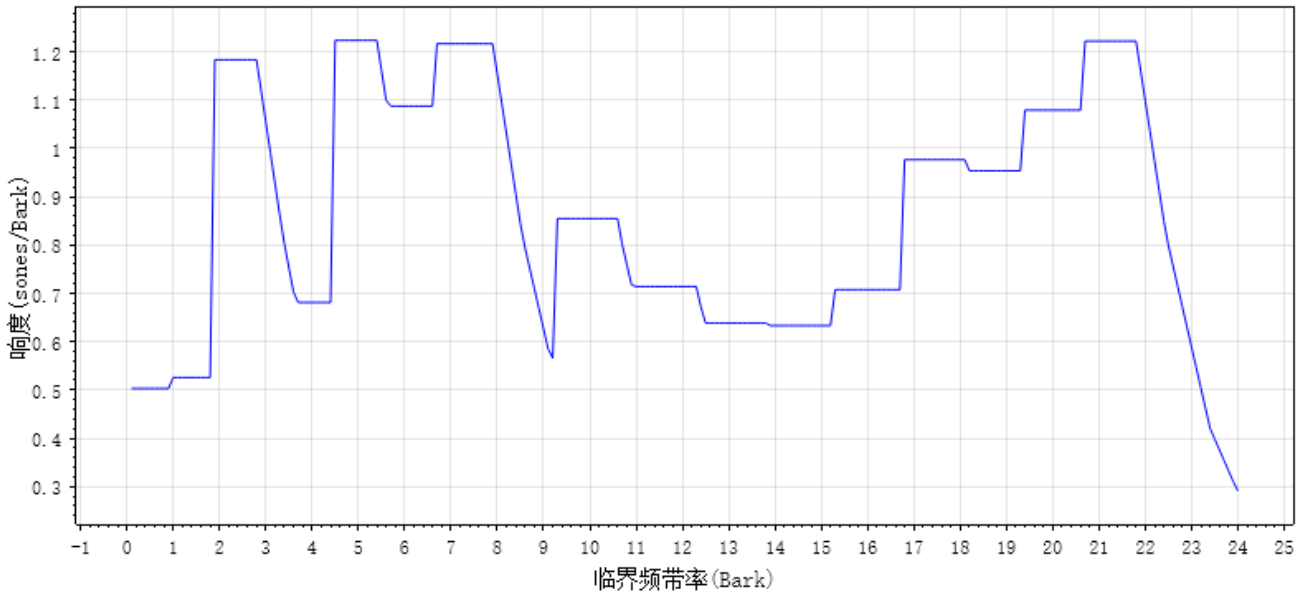
NVH工况1-时变响度-主轴承Acc2



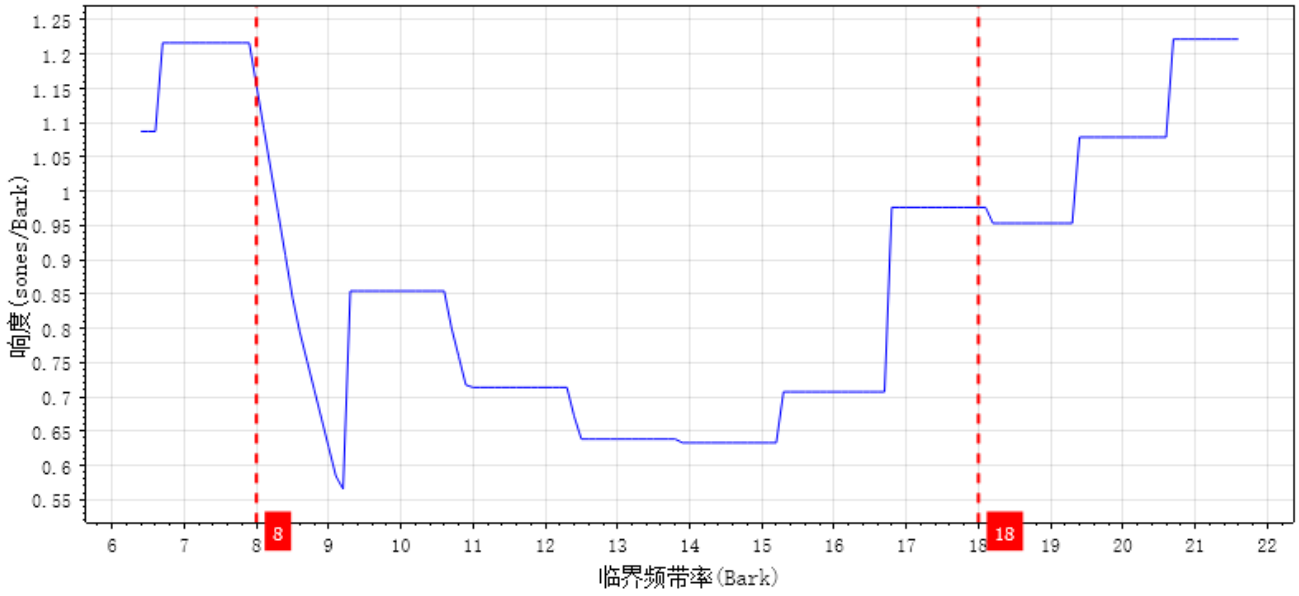
NVH工况1-时变响度-主轴承Acc2-均值(19.96)



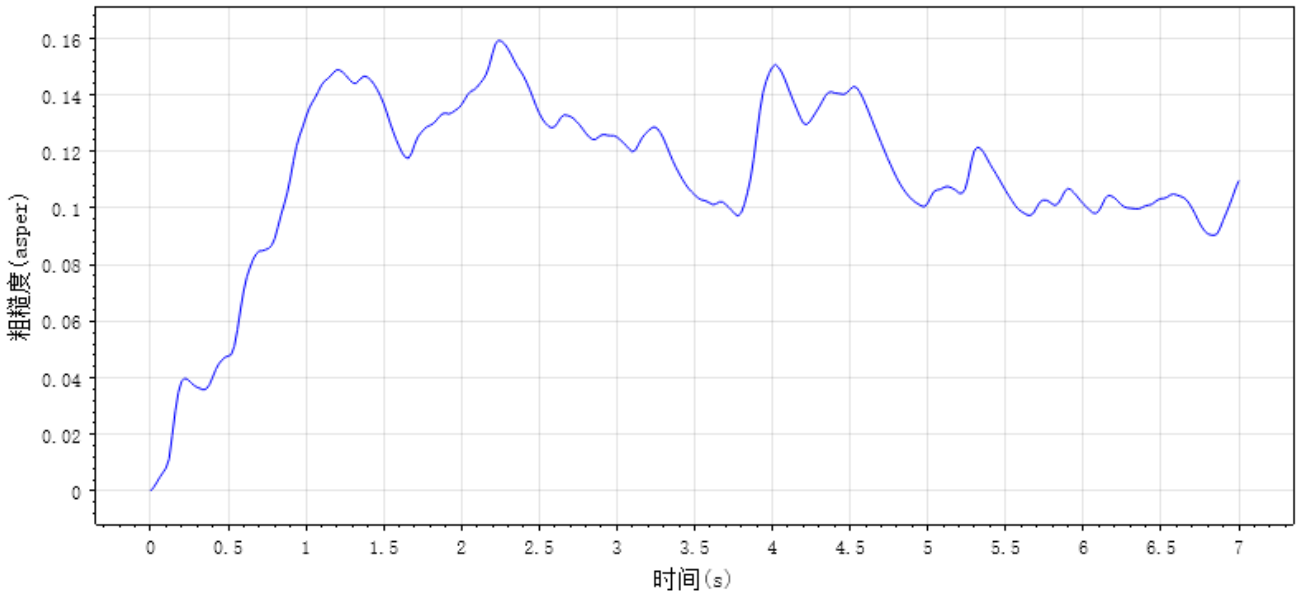
NVH工况1-特征响度-主轴承Acc2



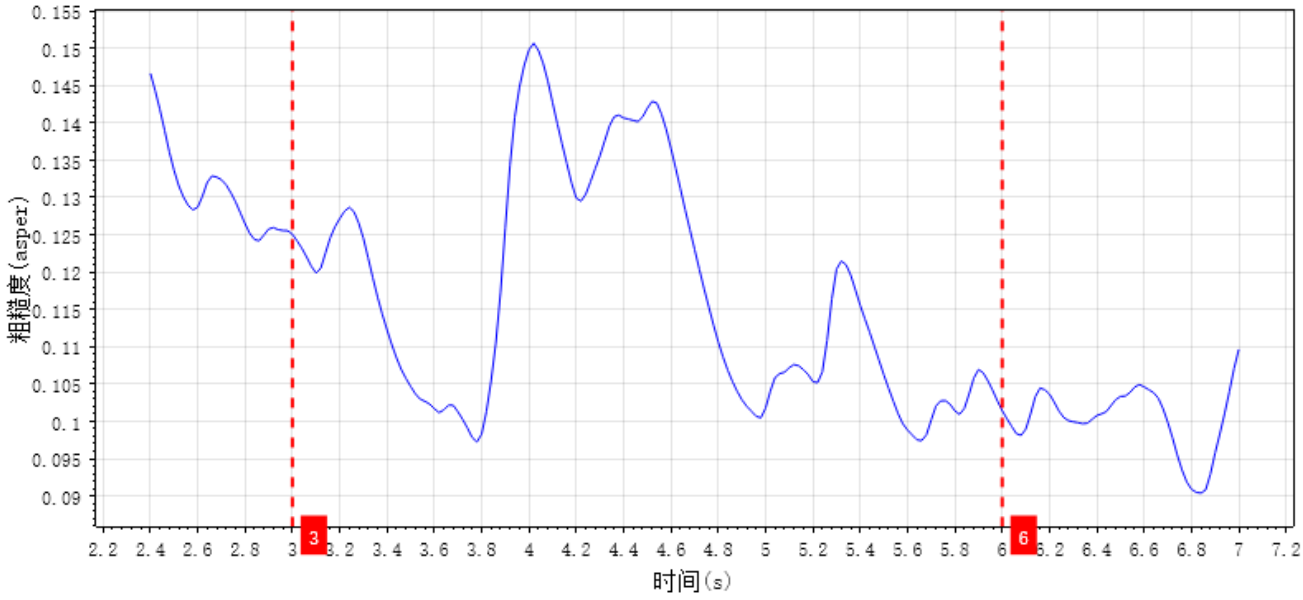
NVH工况1-特征响度-主轴承Acc2-均值(0.76)



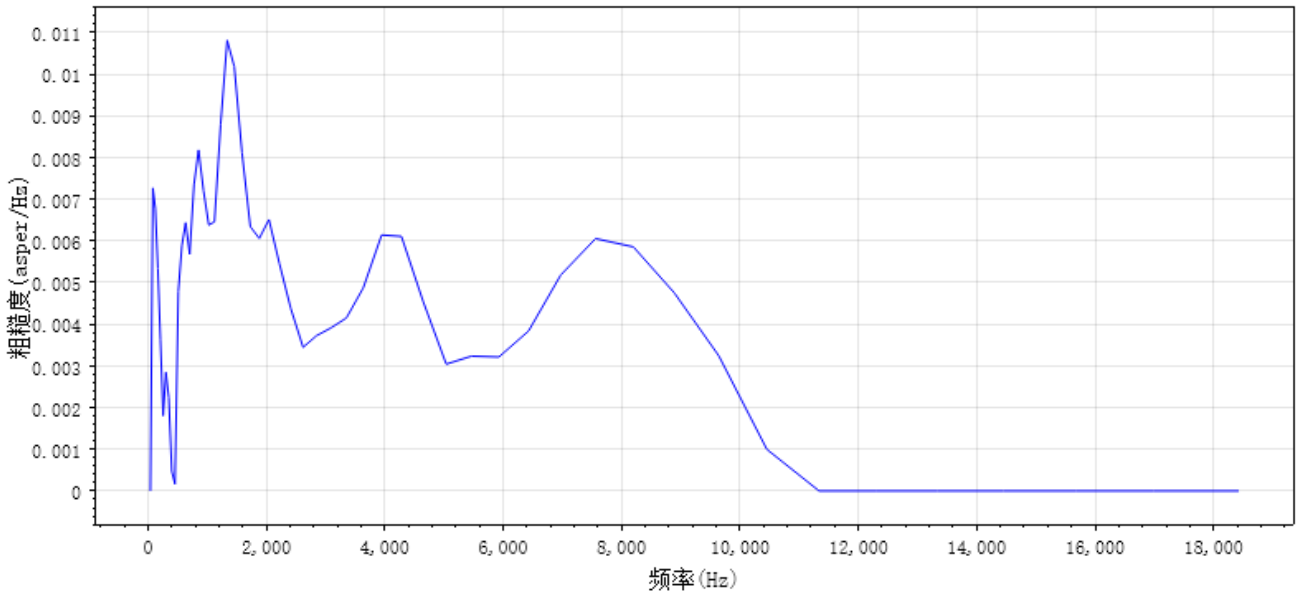
NVH工况1-粗糙度-时间-主轴承Acc2_时变粗糙度_0.1448

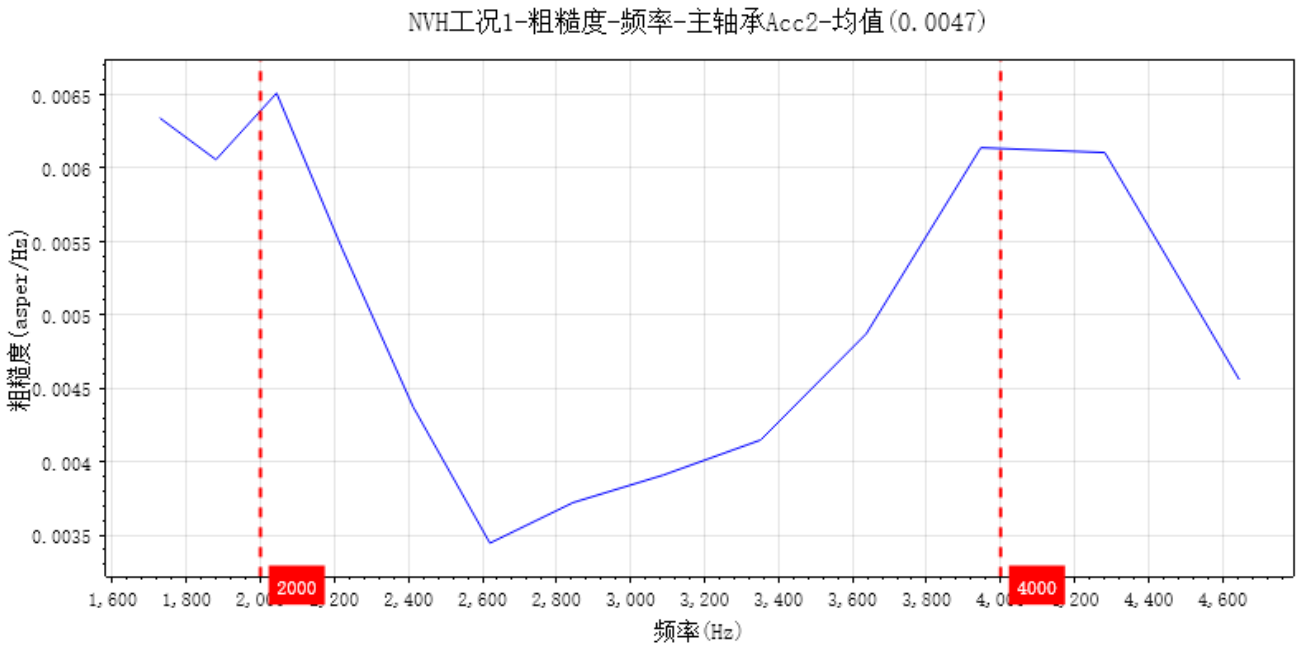


NVH工况1-粗糙度-时间-主轴承Acc2-均值(0.12)



NVH工况1-粗糙度-频率-主轴承Acc2_粗糙度_0.1448

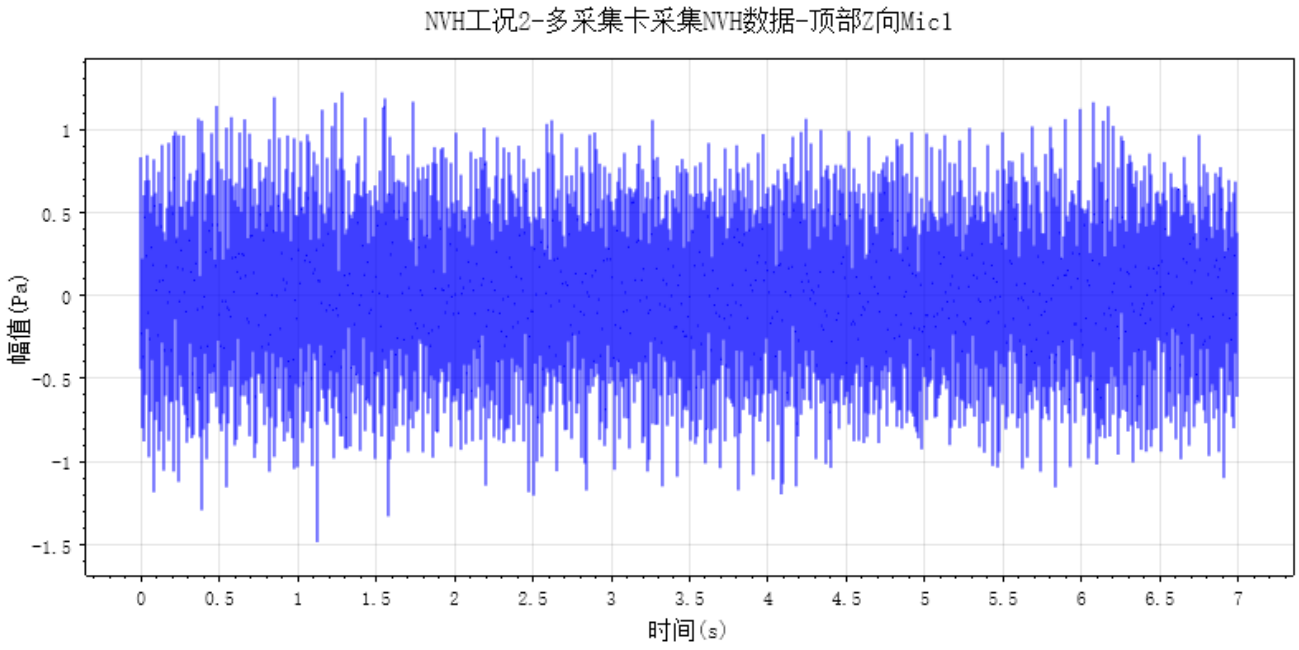




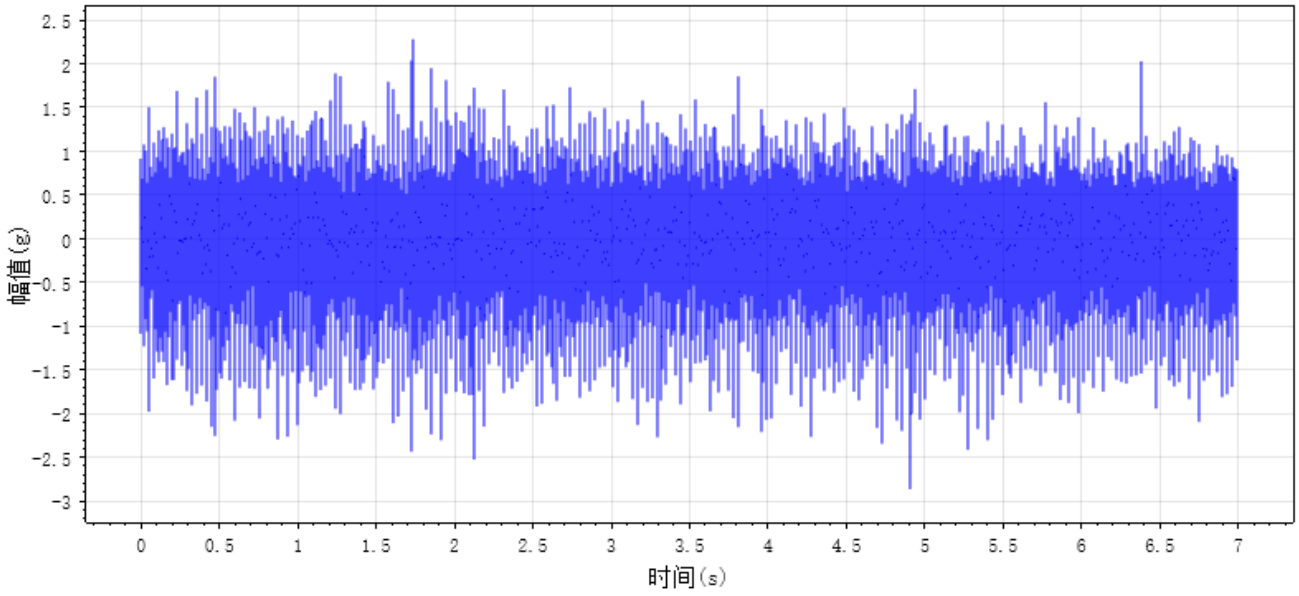
NVH工况2

测试结果：OK

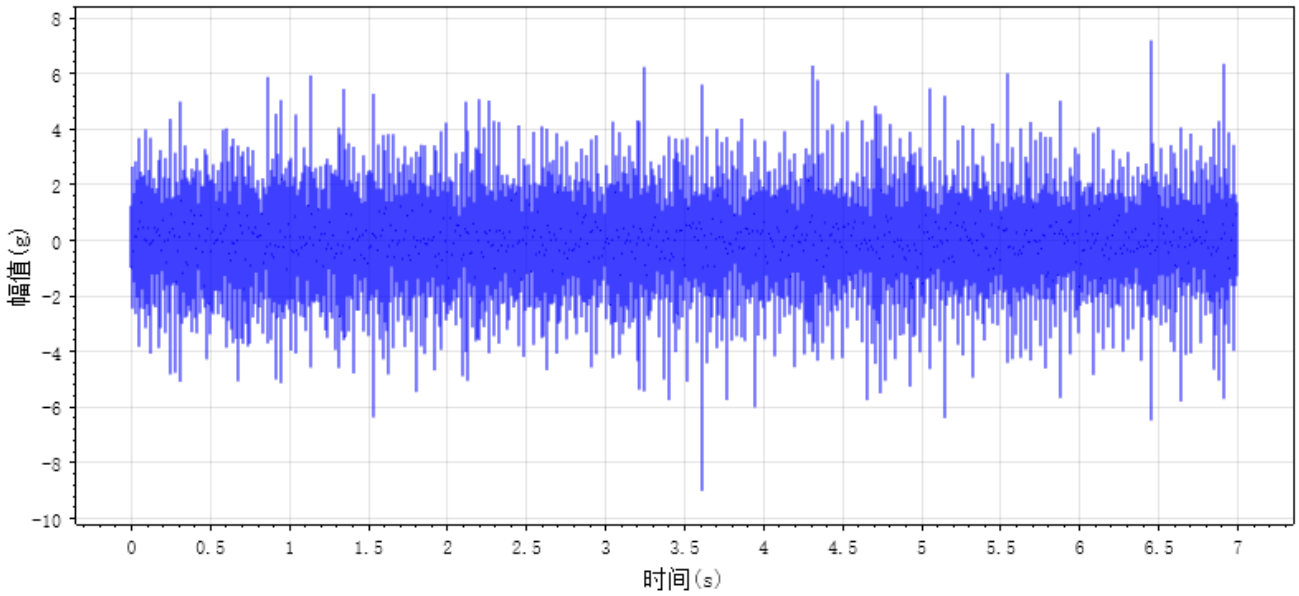
工况信息：转速-2000rpm； 进气压力-0.3MPa； 排气压力-1.5MPa



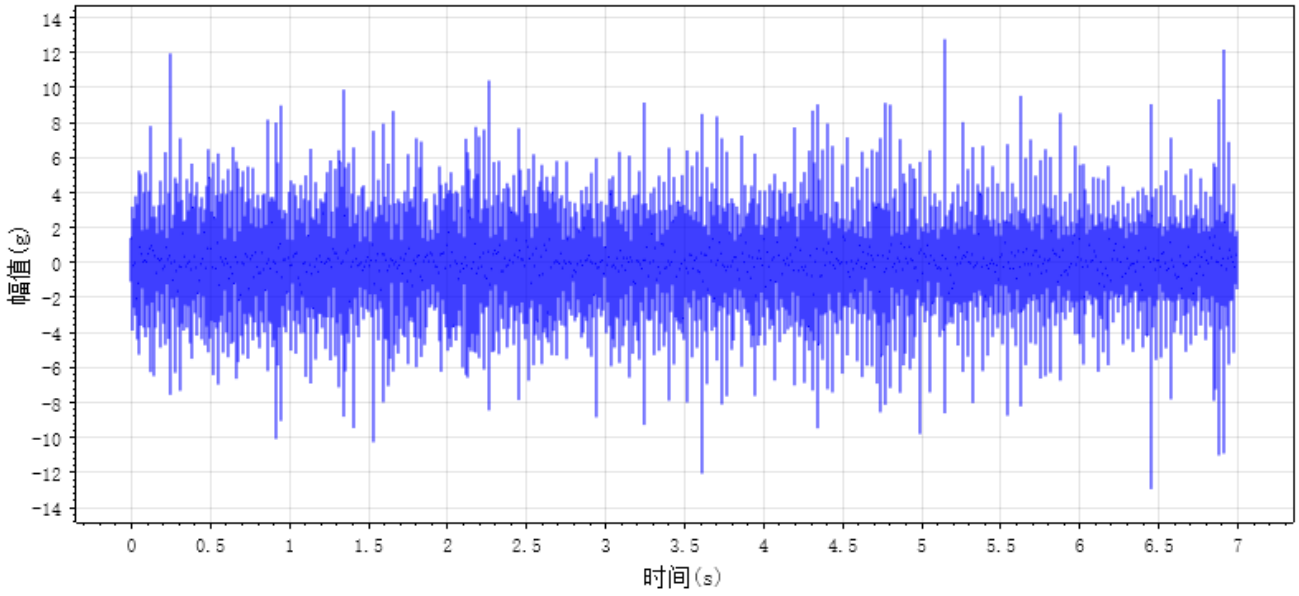
NVH工况2-多采集卡采集NVH数据-副轴承Acc1



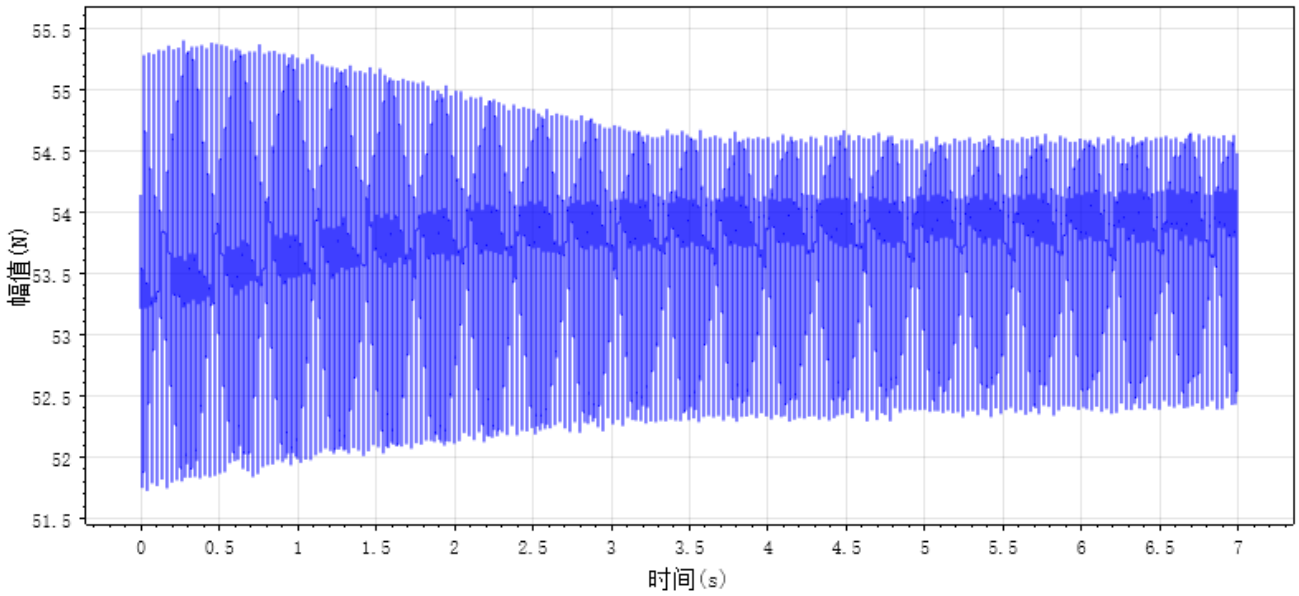
NVH工况2-多采集卡采集NVH数据-主轴承Acc2



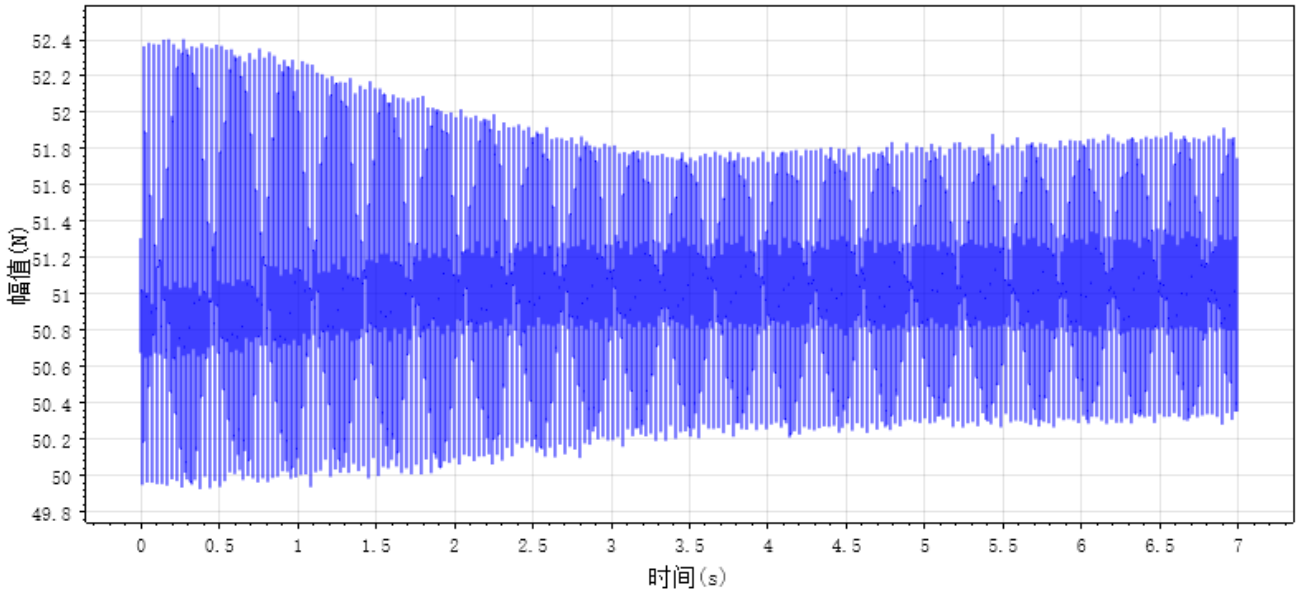
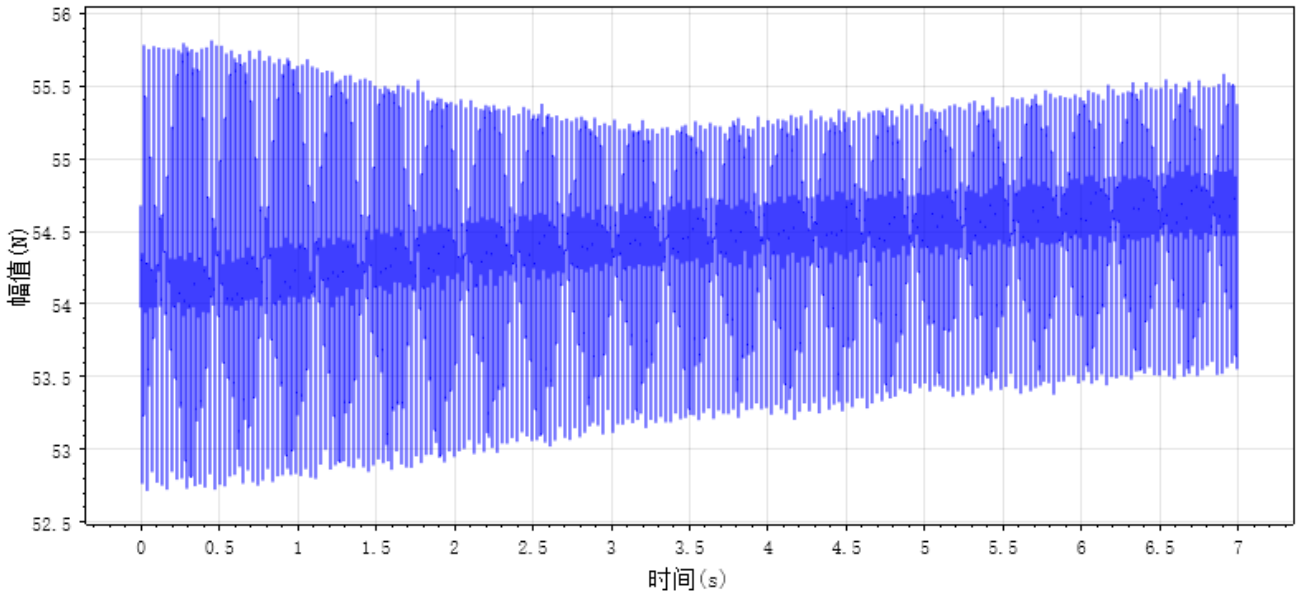
NVH工况2-多采集卡采集NVH数据-涡旋盘Acc3



NVH工况2-多采集卡采集NVH数据-SS824-力馈1



NVH工况2-多采集卡采集NVH数据-SS824-力馈2



NVH工况2-多采集卡采集NVH数据-SS824-力馈3